

Hands-on Activity 1.7	
2D Essentials & Publishing Essentials	
Course Code: CPE 411	Program: Computer Engineering
Course Title: Interactive Extended Reality	Date Performed: 31/01/2026
Section: 22S2	Date Submitted: 31/01/2026
Name(s): Ross Andrew Bulaong	Instructor: Engr. Jimlord M. Quejado

1. Discussion

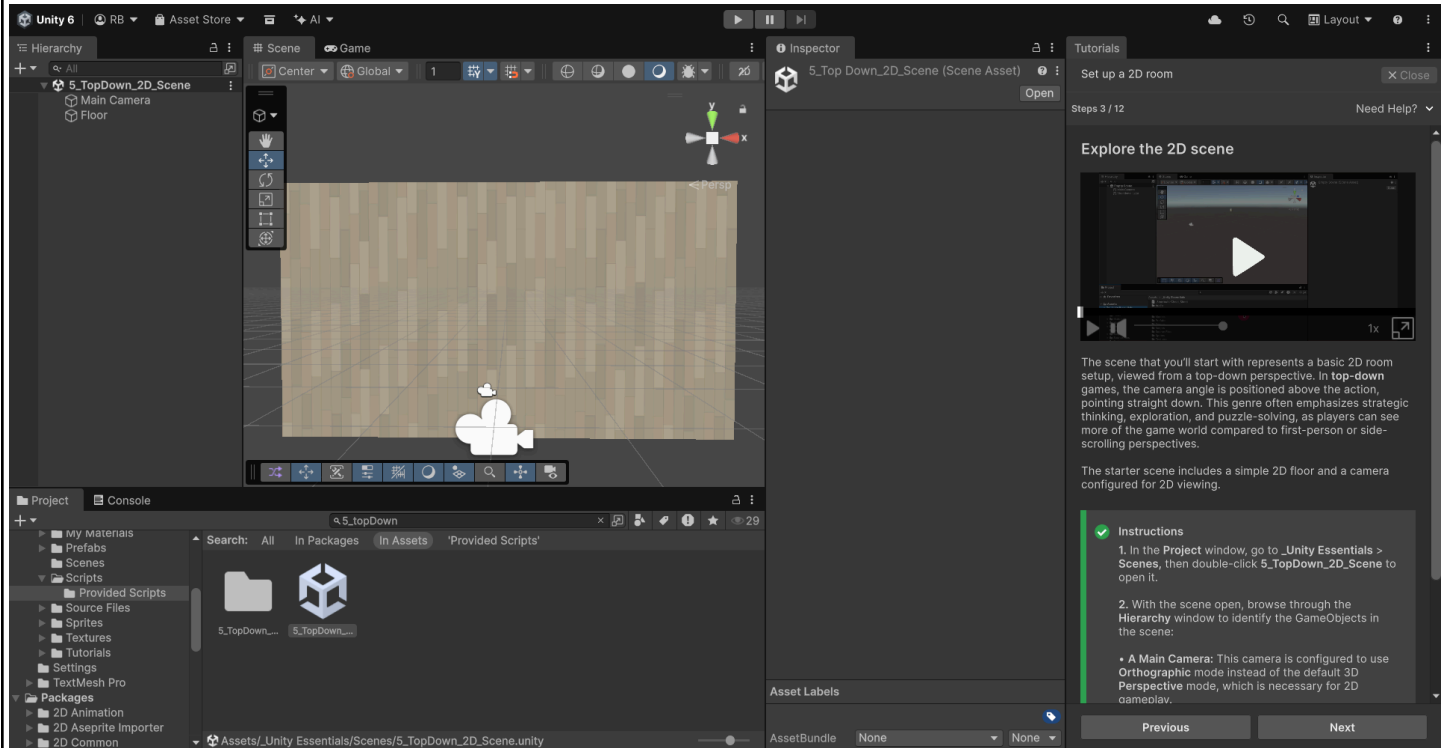
This activity tutors the user in the key aspects that one must learn in order to navigate and/or create 2D games. This activity also teaches the user how they can upload their creations to WebGL / UnityPlay.

2. Materials and Equipment

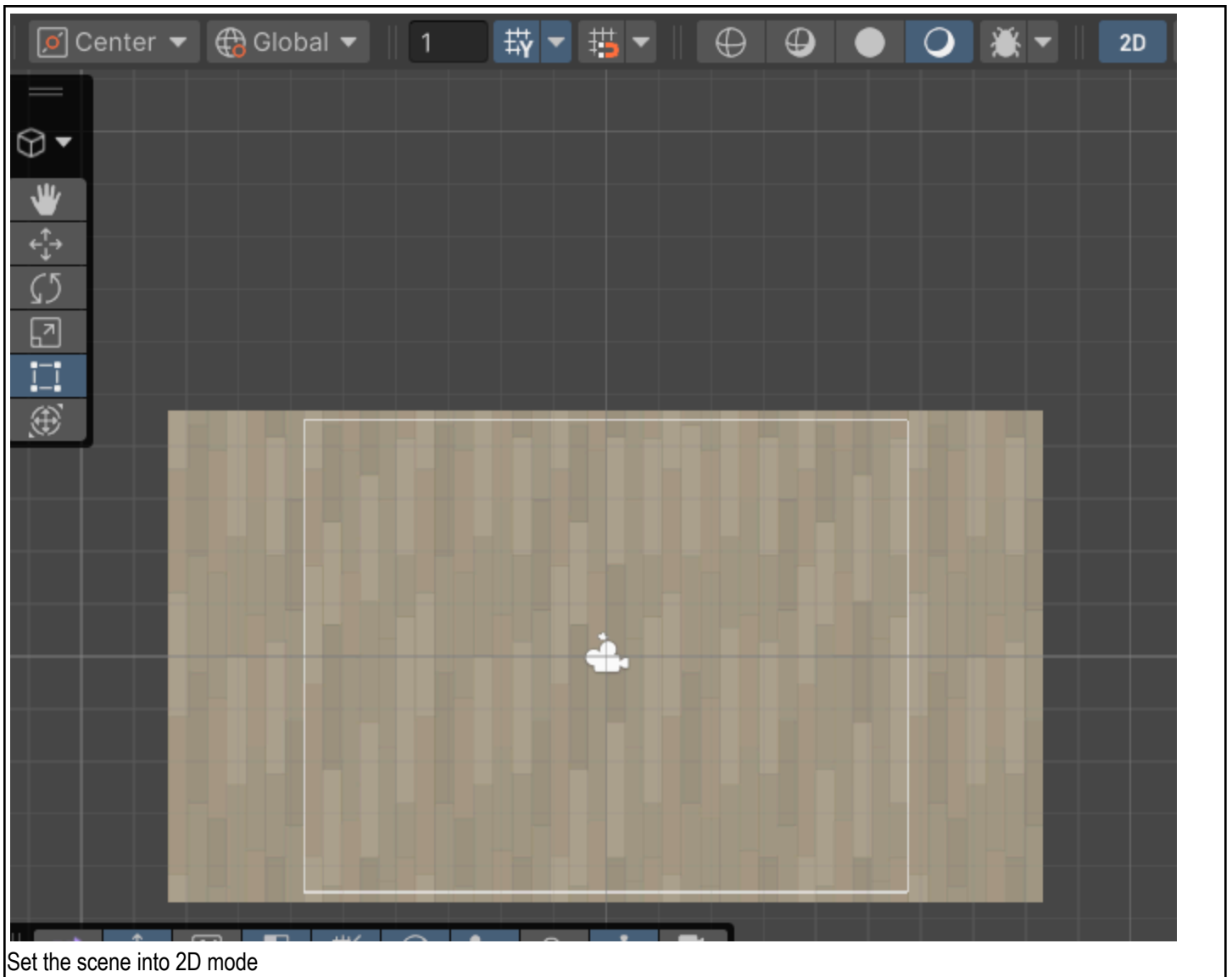
Unity Hub and Unity Editor
Learn Unity Website ([Unity Essentials Pathway - Learn Game Development for Beginners | Unity Learn](#))
Computer with Network Connection

3. Procedure

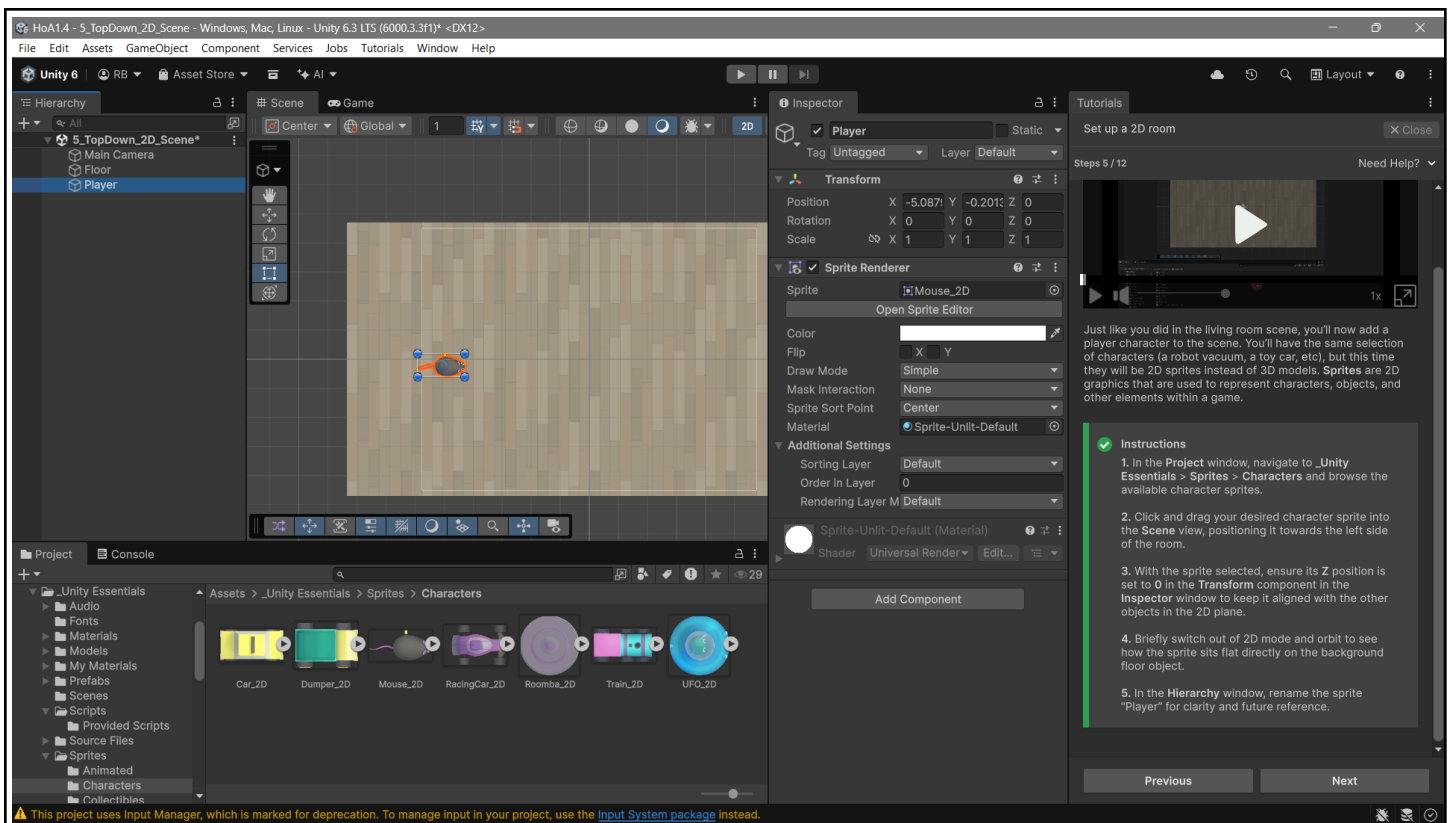
2D ESSENTIALS



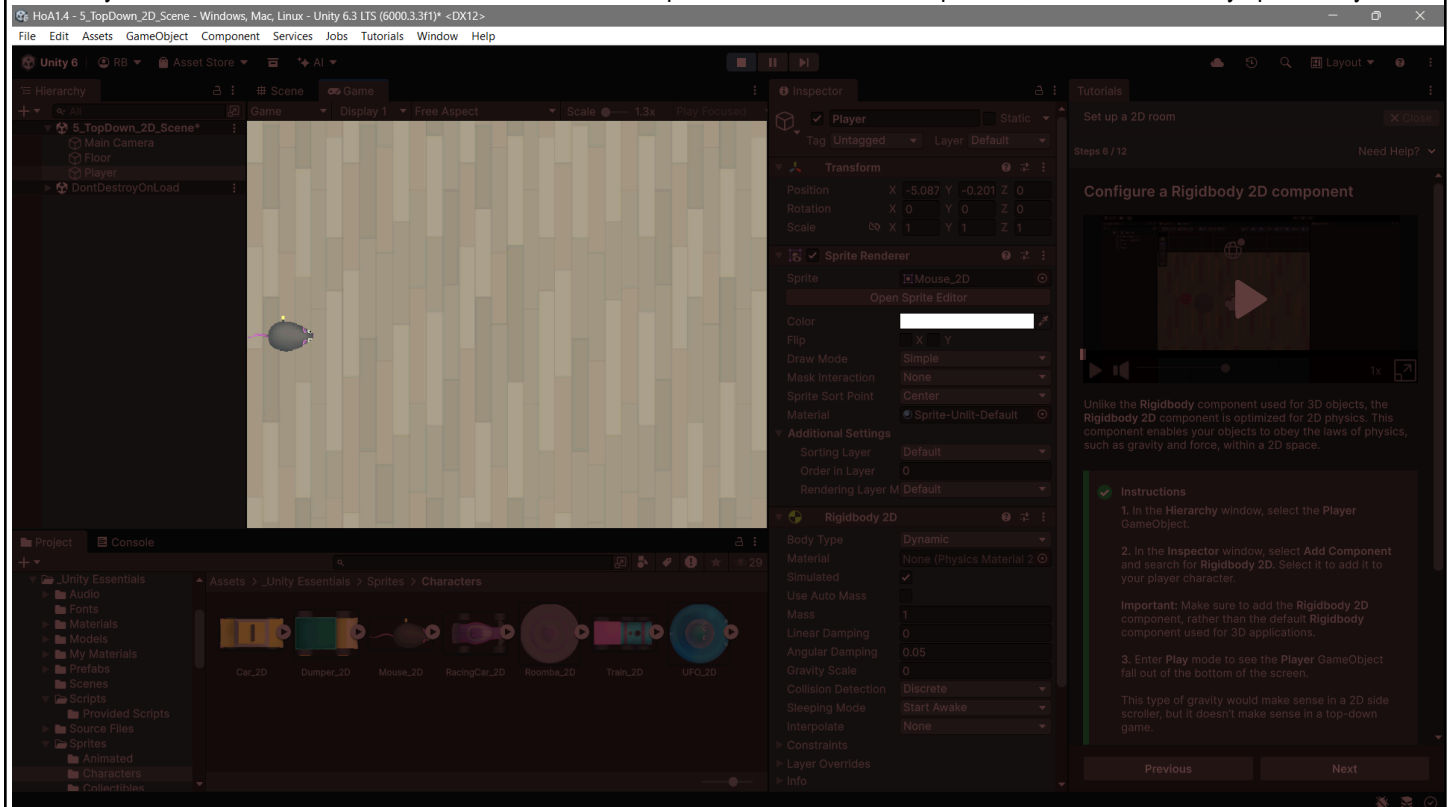
Exploring the 2d scene, which appears to have a 2D wall object serving as the background



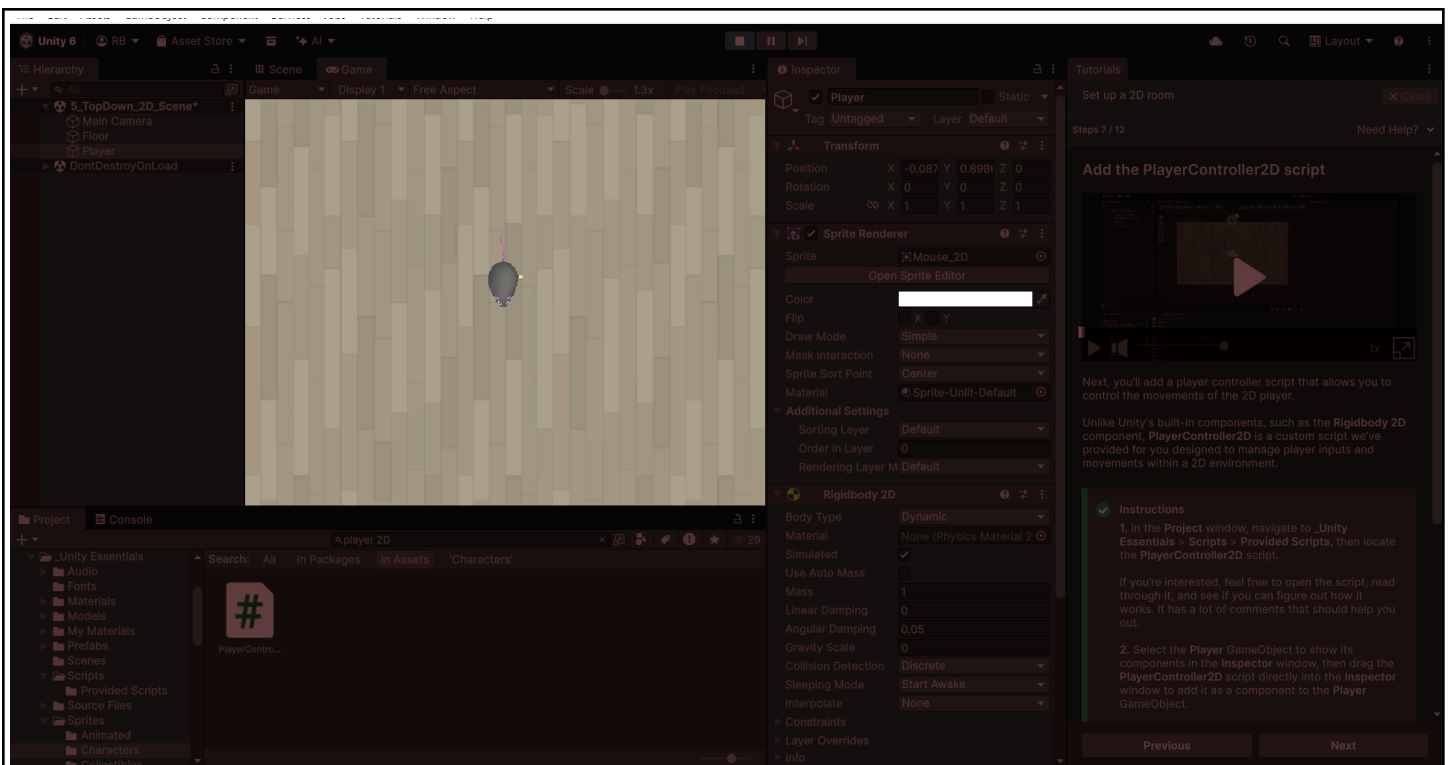
Set the scene into 2D mode



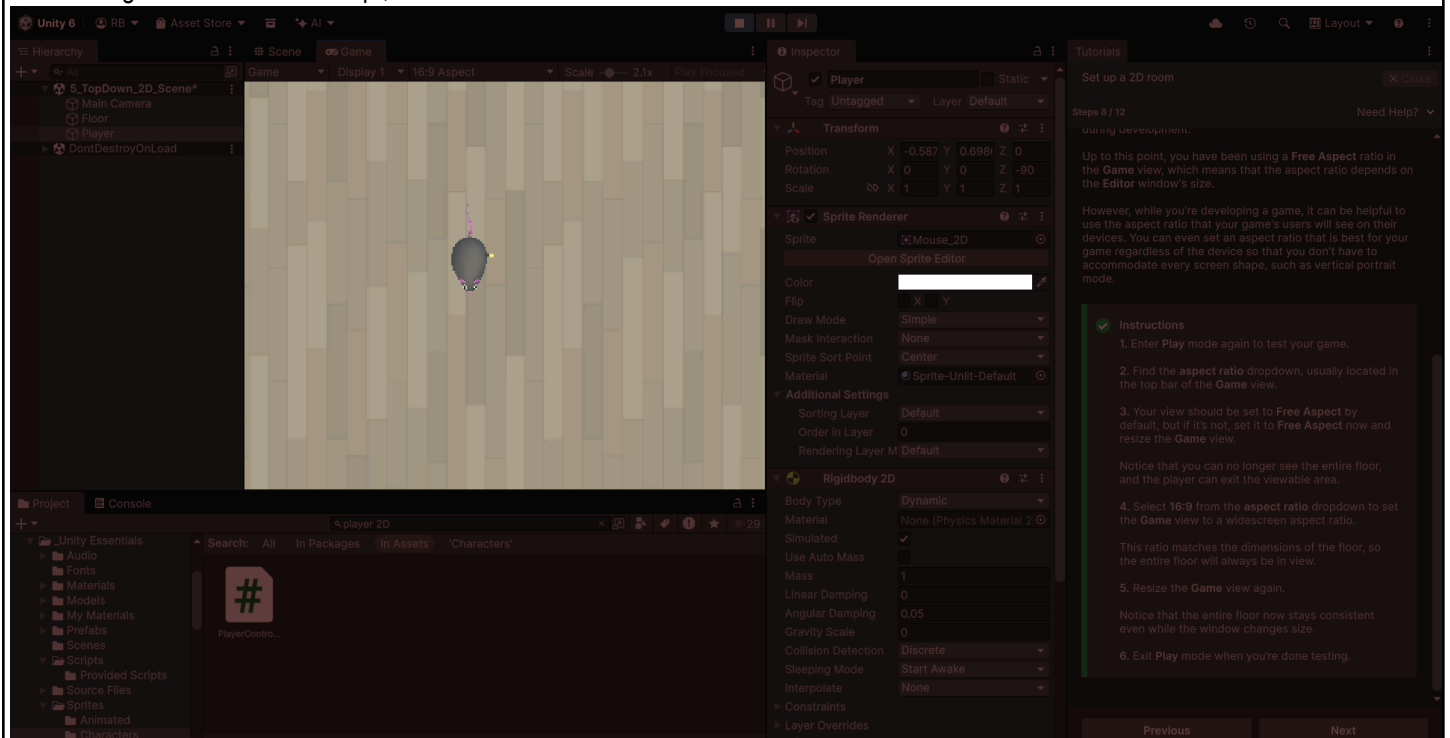
Added my mouse character to the 2D scene; checked if the Z position is zero to ensure pure 2D, i then renamed my sprite "Player"



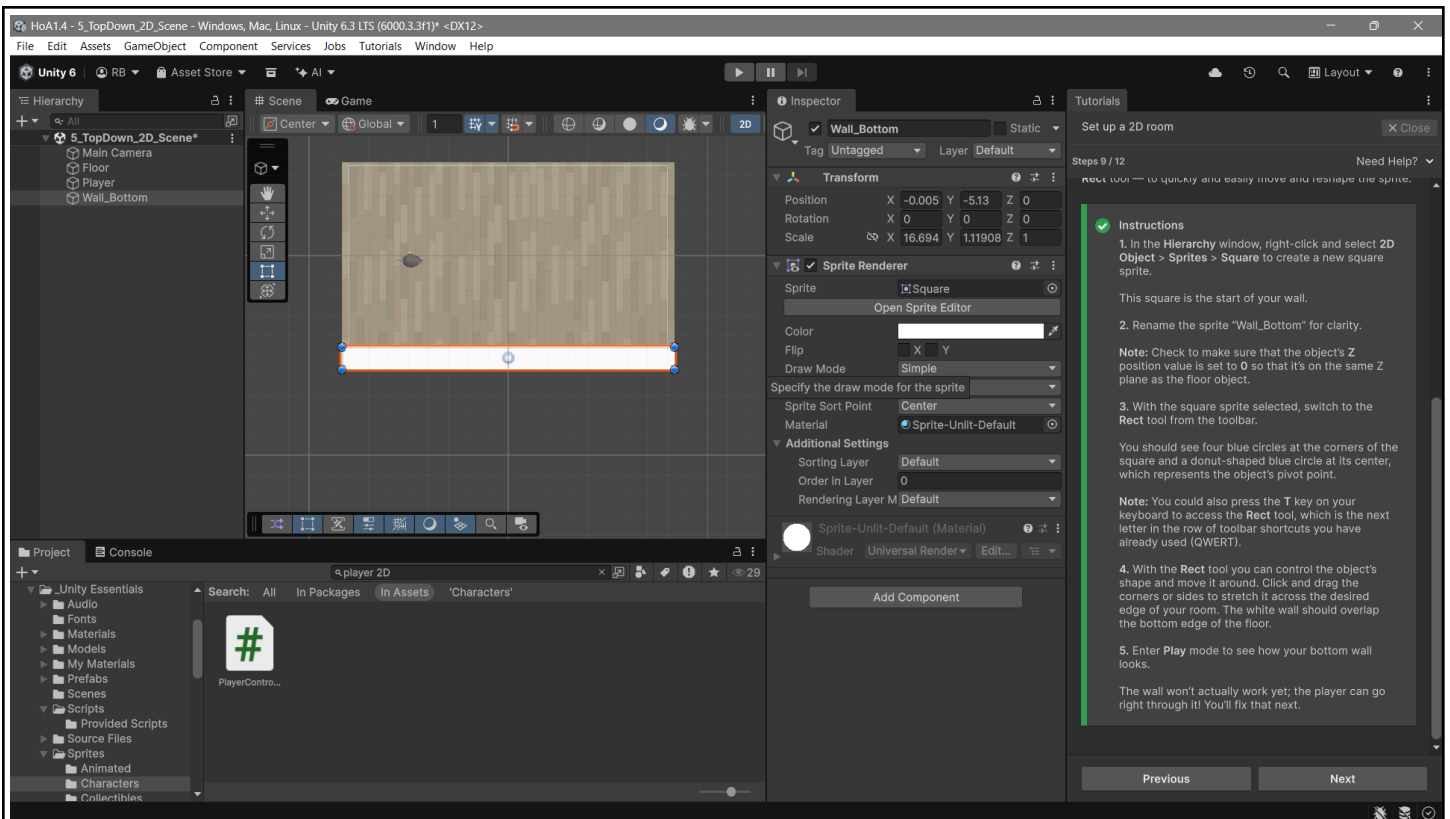
Added a 2D rigidbody component, set the gravity to zero so that the mouse doesn't fall off the screen



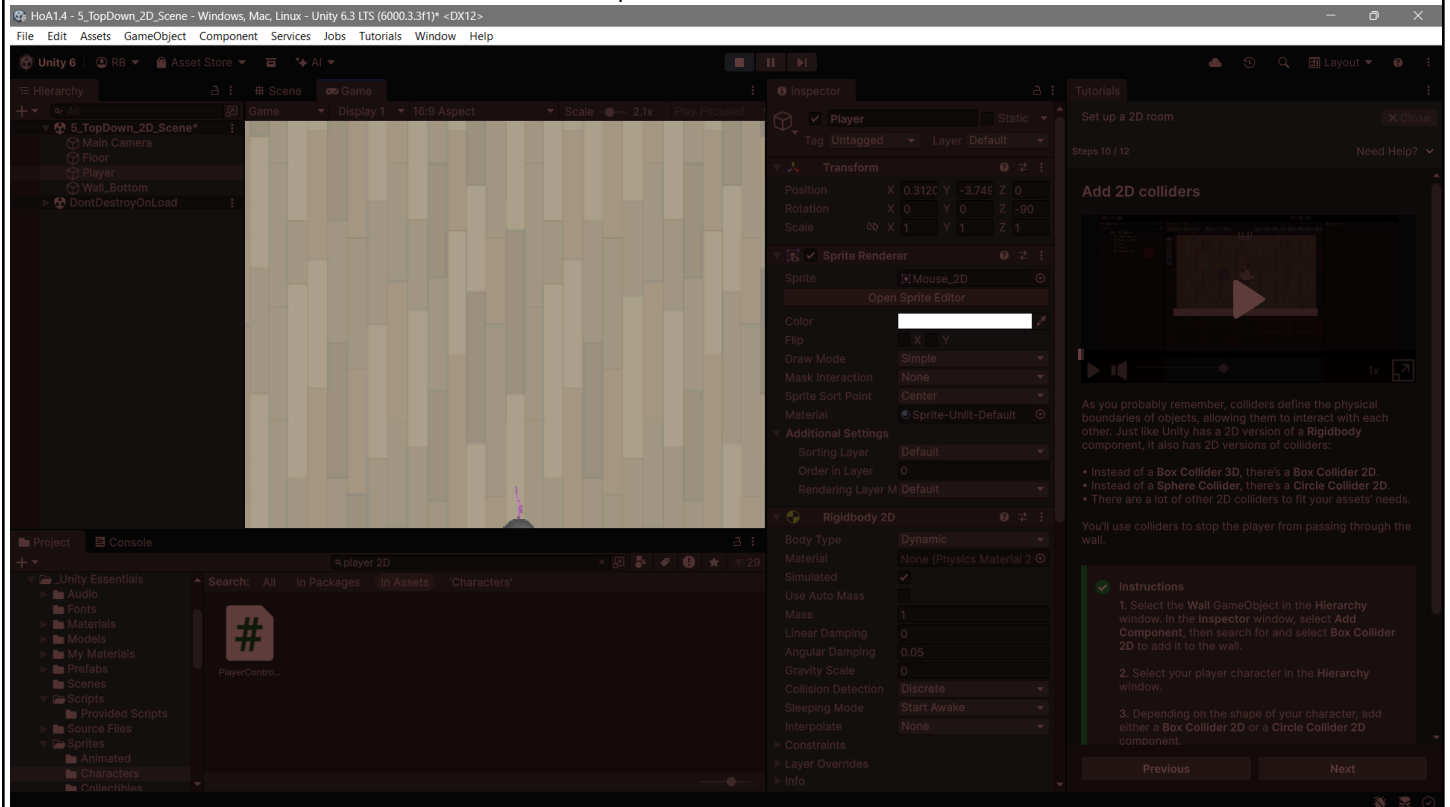
After adding the 2D controller script, I am now able to move the mouse as i see fit



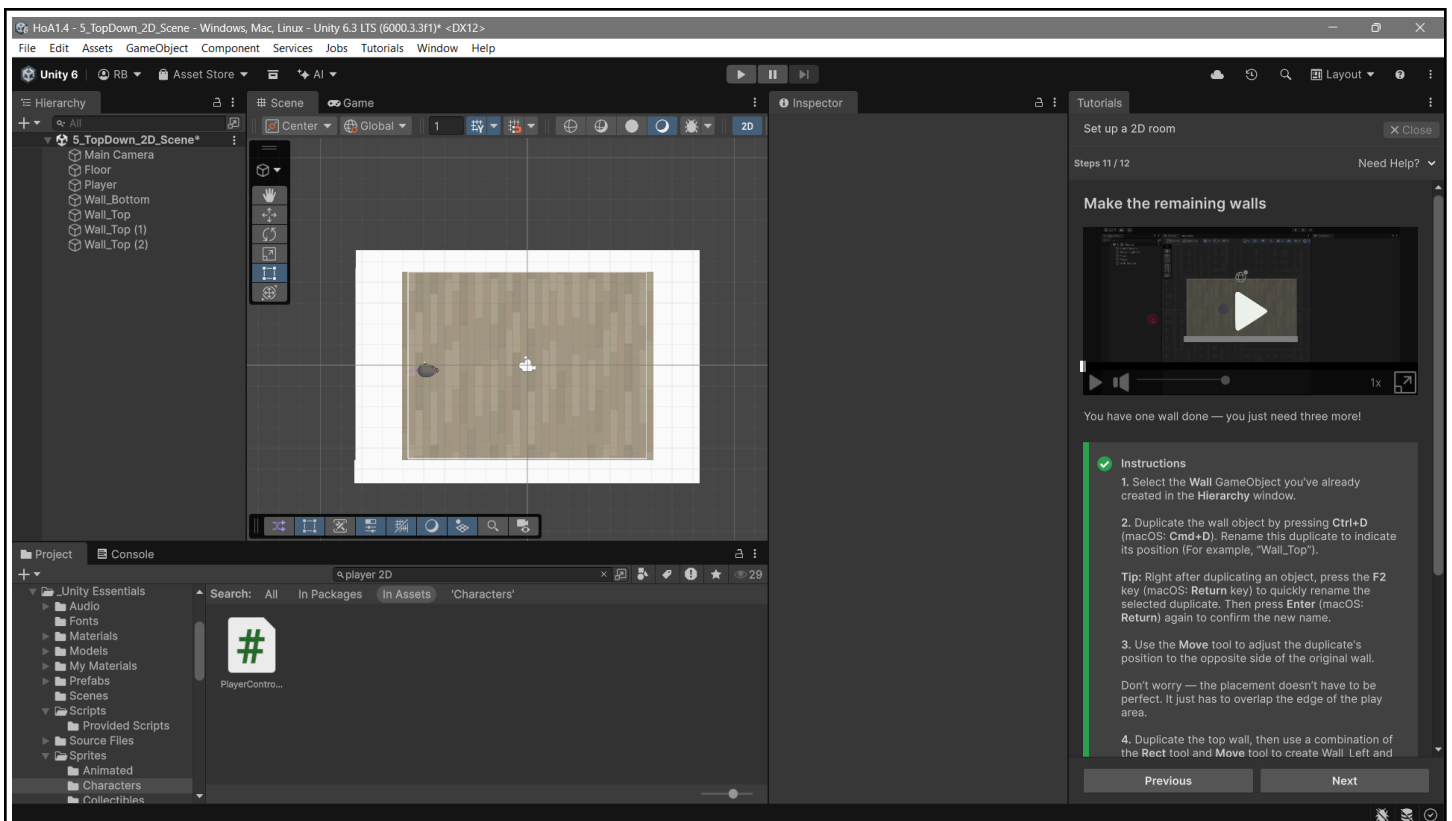
I fiddled around the Aspect Ratio, then set it to 16:9 as instructed



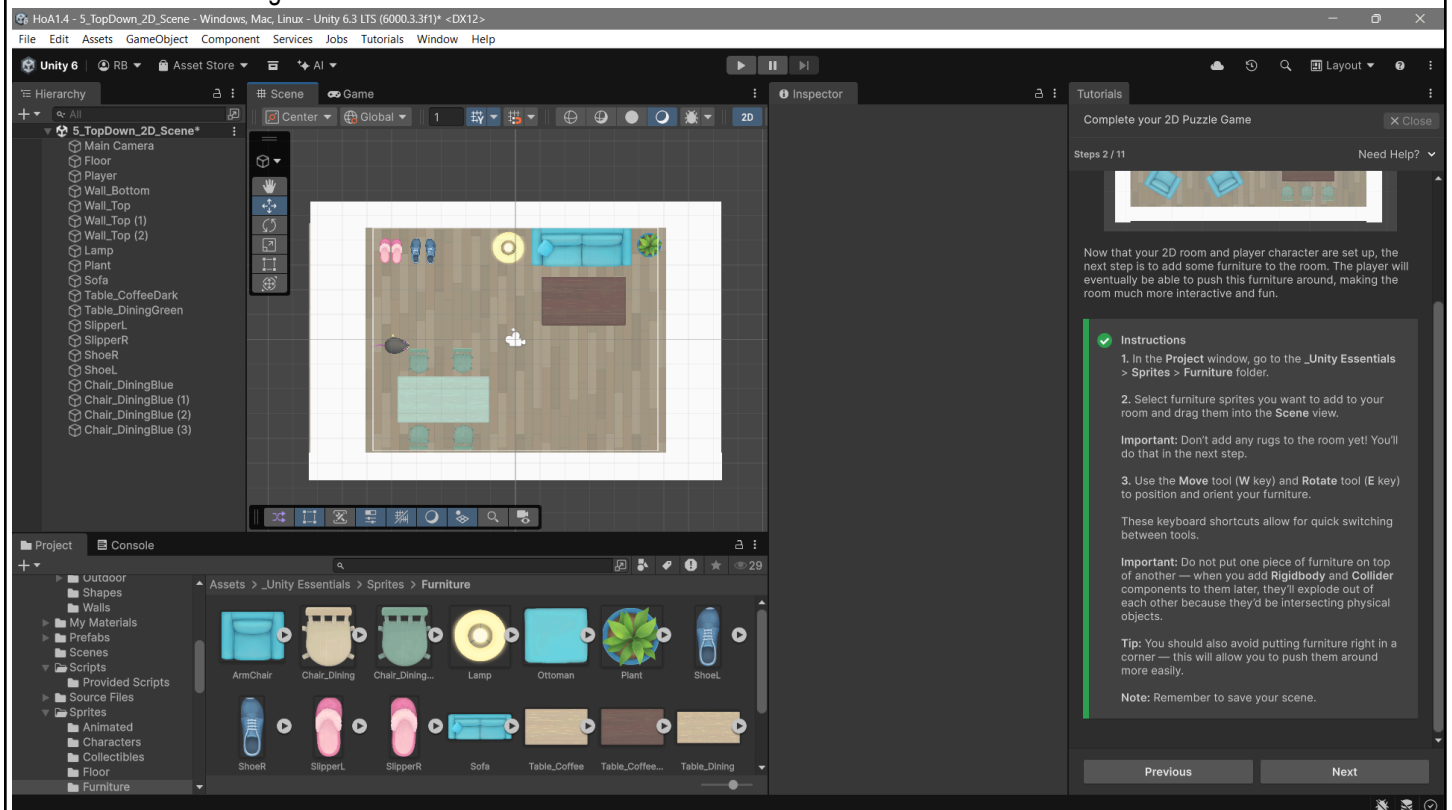
Added a white bottom wall and used the rect tool to reshape it as instructed



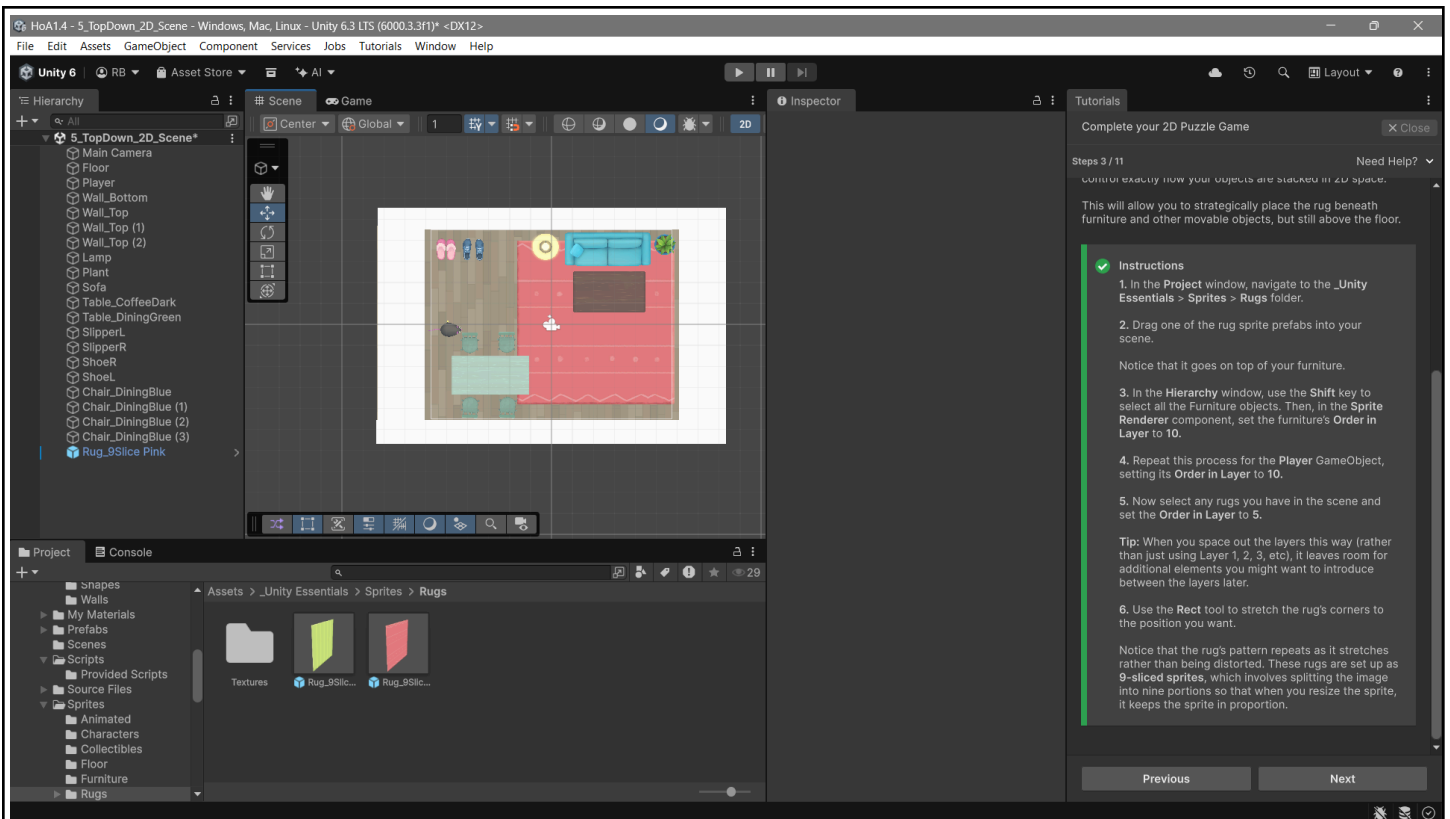
After adding A collider to my mouse as well as to the wall, the mouse now doesn't go off screen and instead just hits the wall.



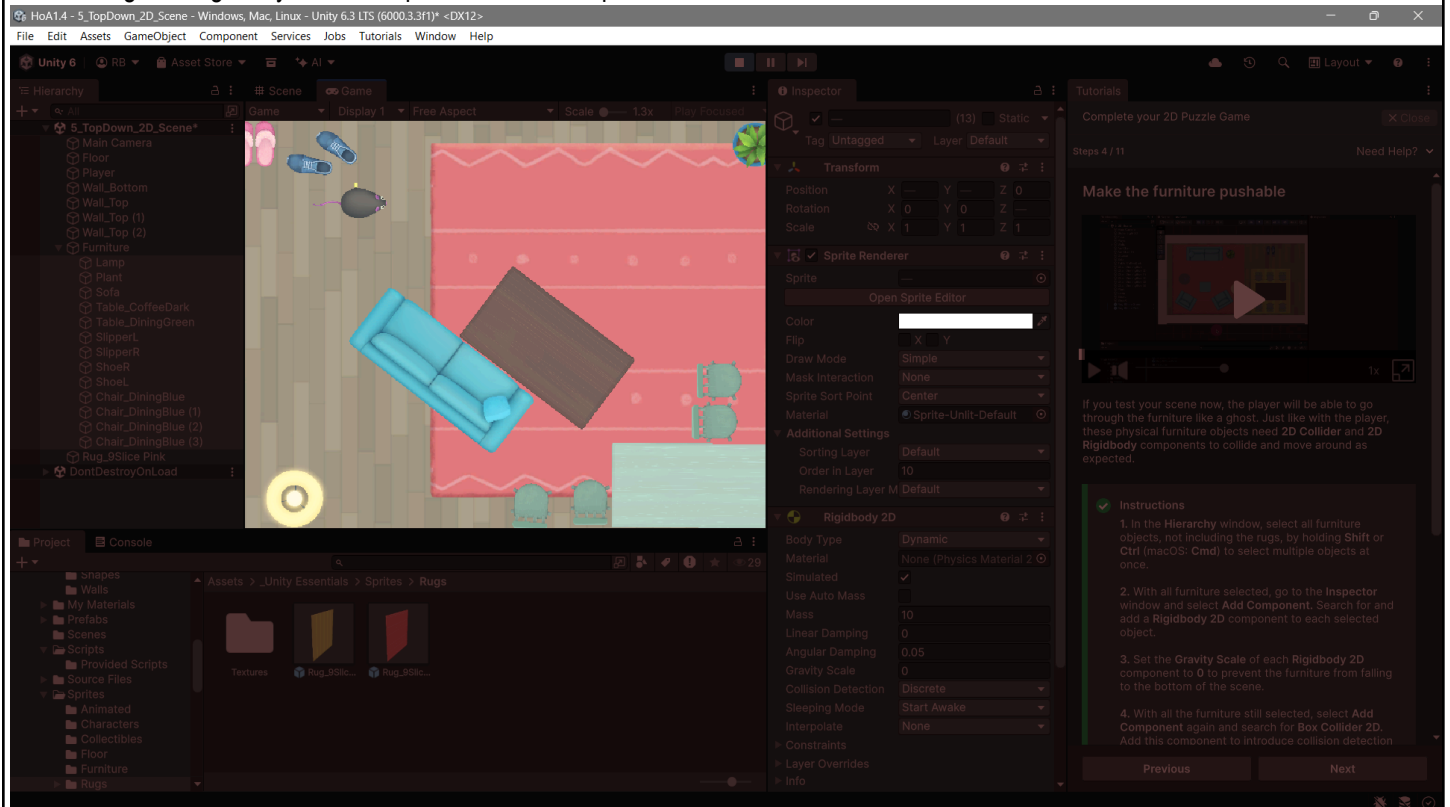
Made the other remaining walls



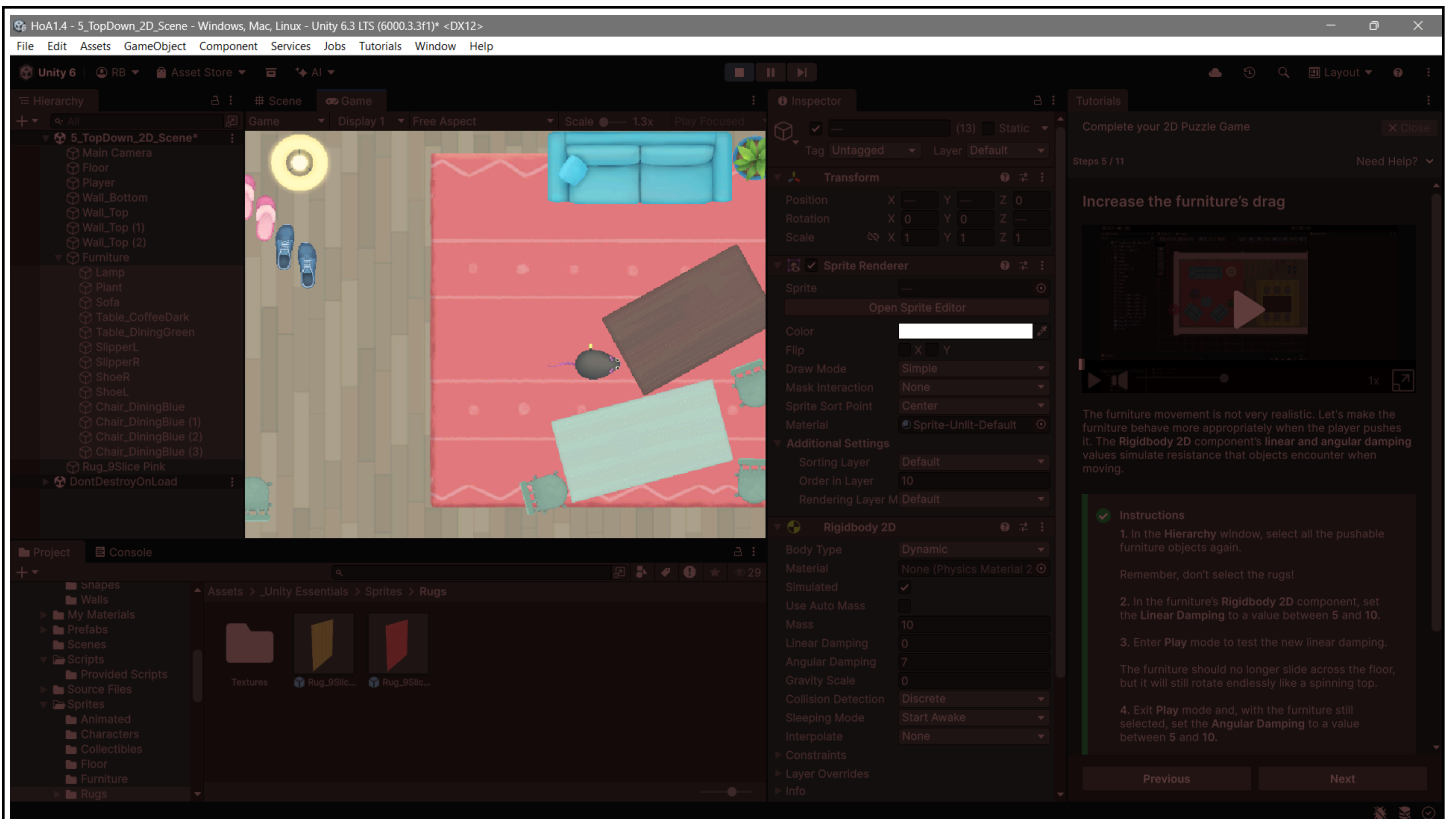
Added furniture to the 2D space as instructed



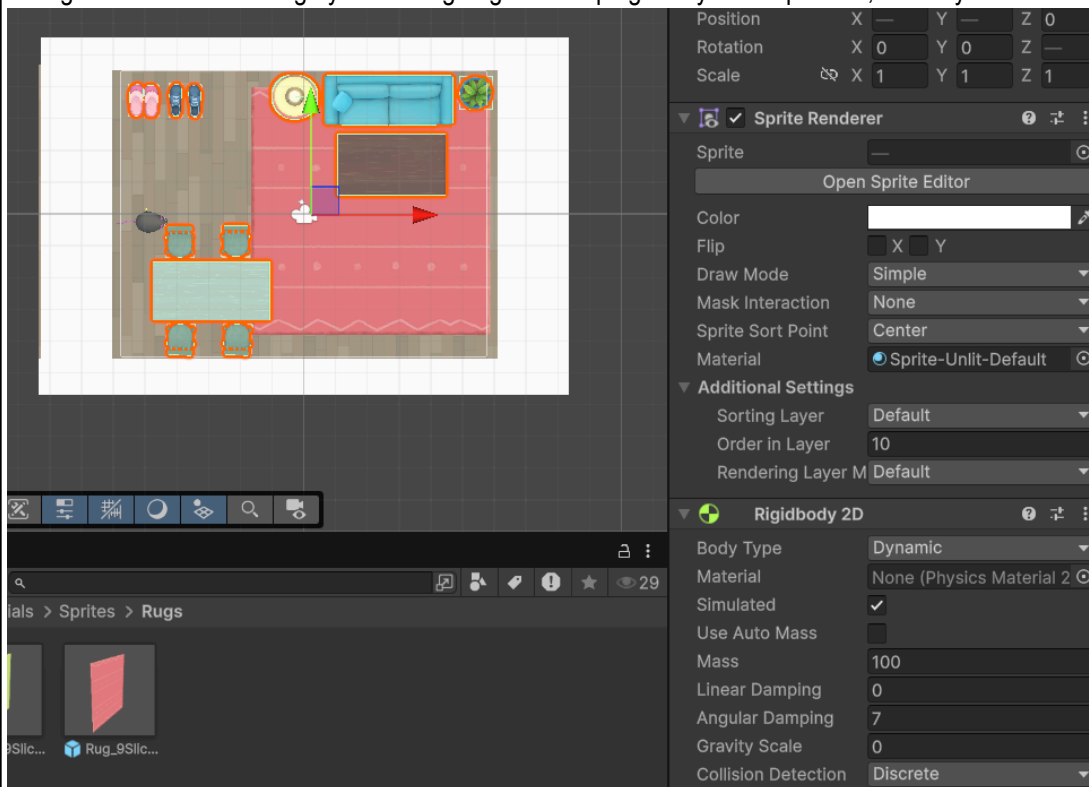
Added a rug, managed layers in the sprite renderer component



I've made the furniture pushable by adding either box or circle colliders and 2D rigid bodies to each furniture, now making them interactable / pushable



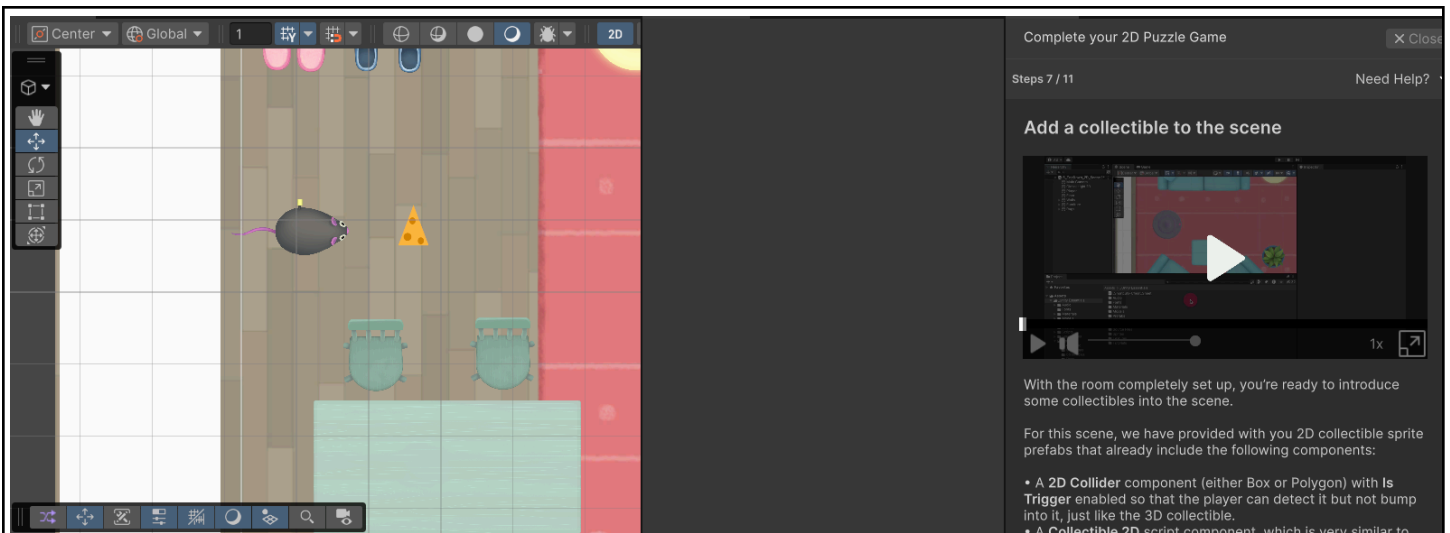
Changed the furnitures' drag by increasing angular damping. They don't spin now, but they still slide off when being pushed



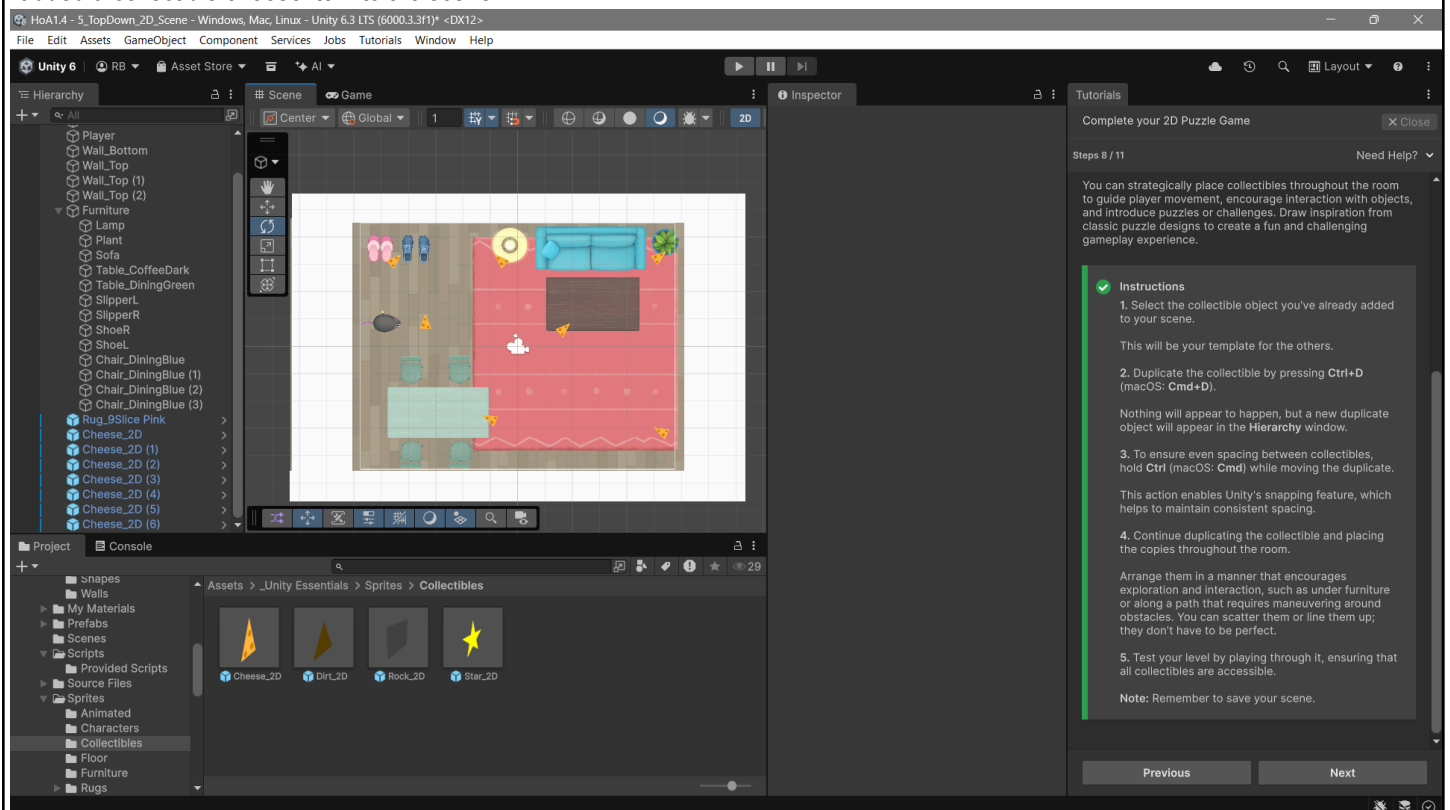
I changed each furniture's mass to 100. No tiny mouse should be able to push objects 100x its weight normally



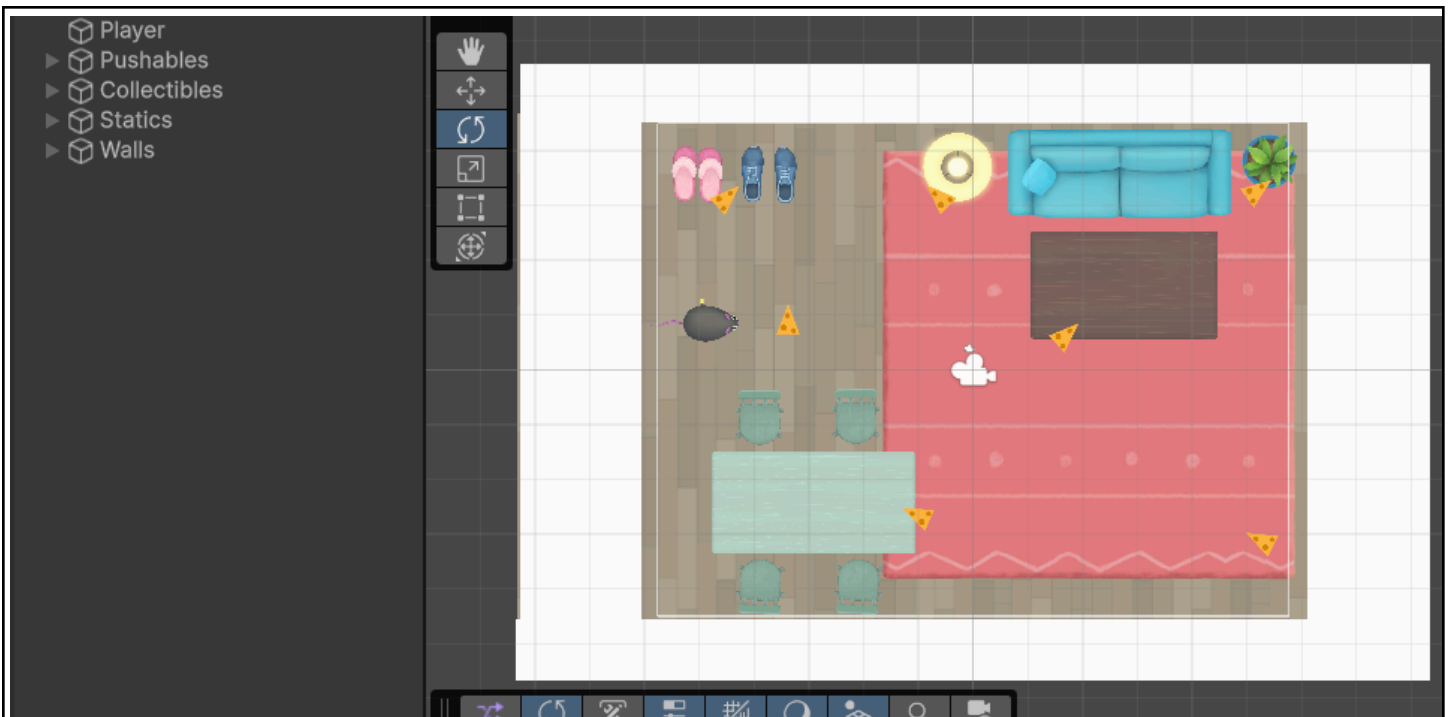
I then tinkered with it again as I feel like the 100 mass is not fun and is rather annoying to play



I added a collectible cheese item to the scene



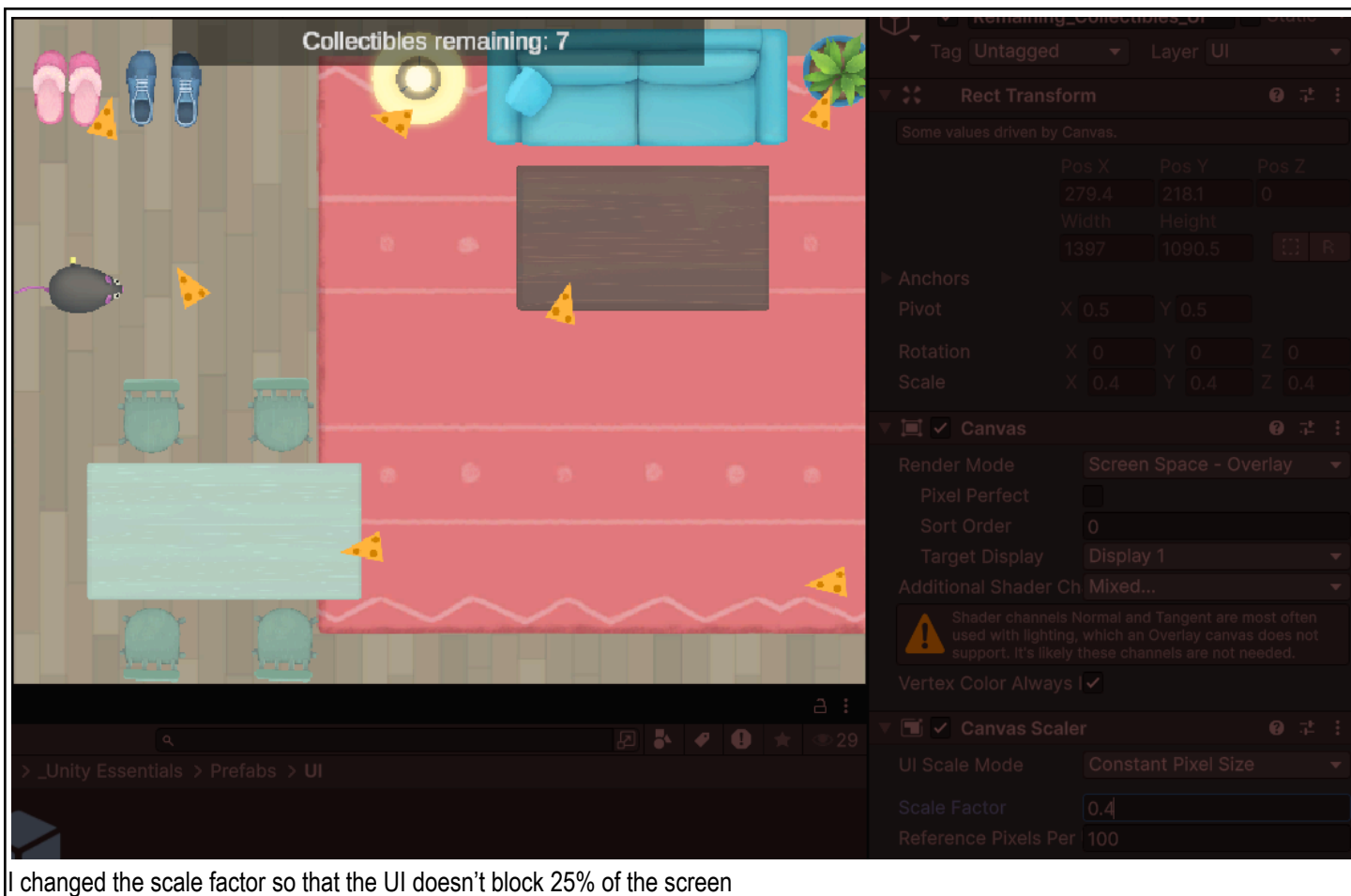
I added more collectibles in a manner that requires the user to explore the whole room / push objects to obtain

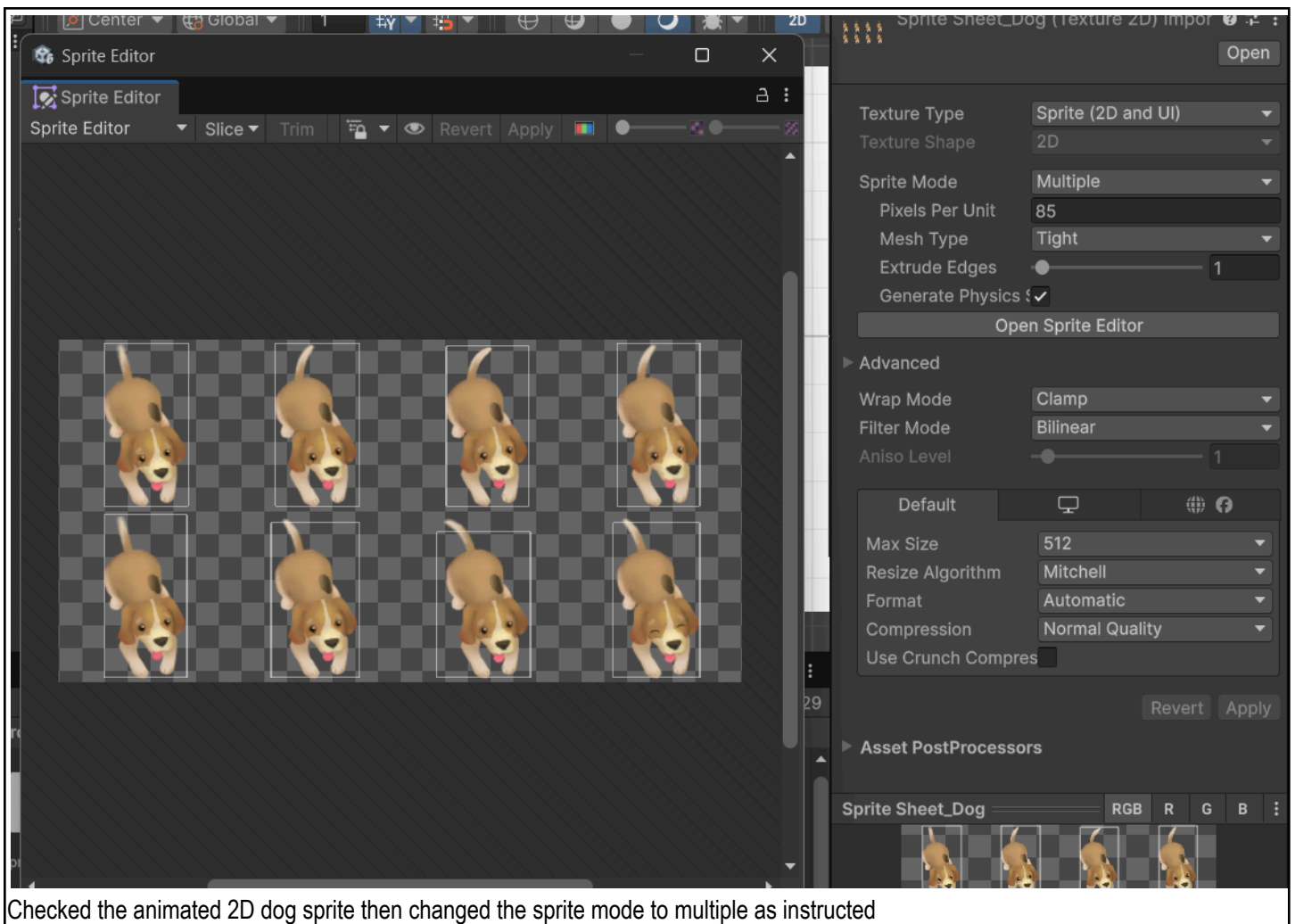


Cleaned up the hierarchy window by properly grouping everything

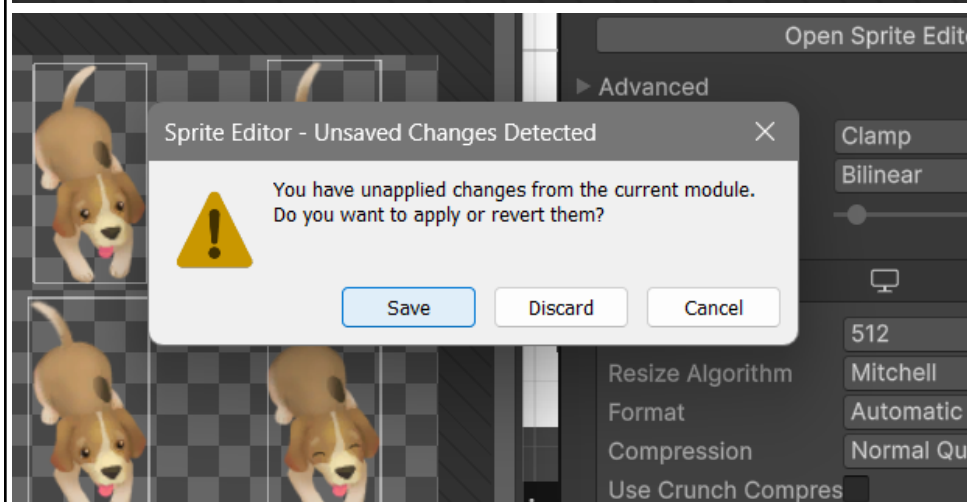
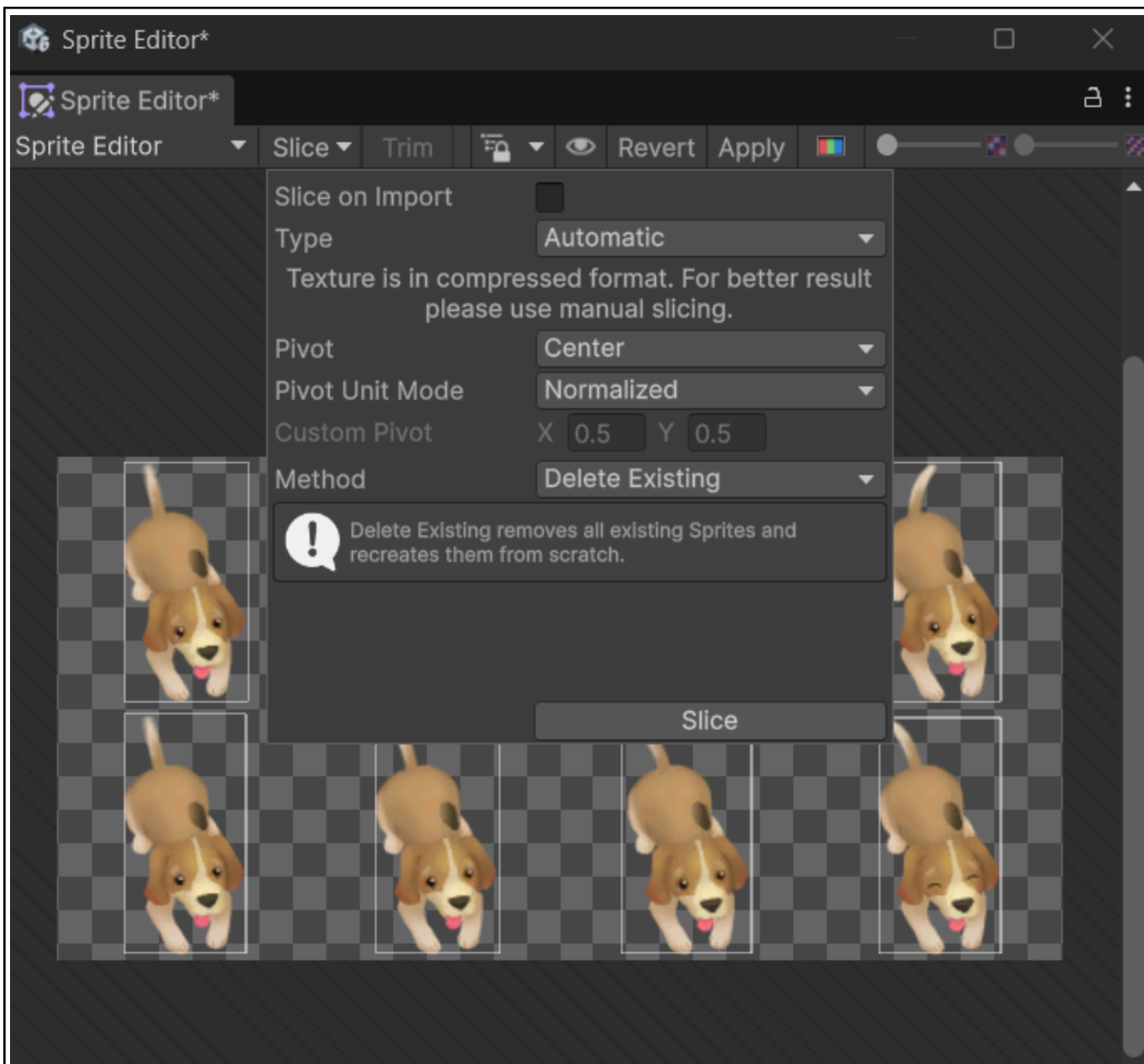


Added the UI prefab that shows how many collectibles remain

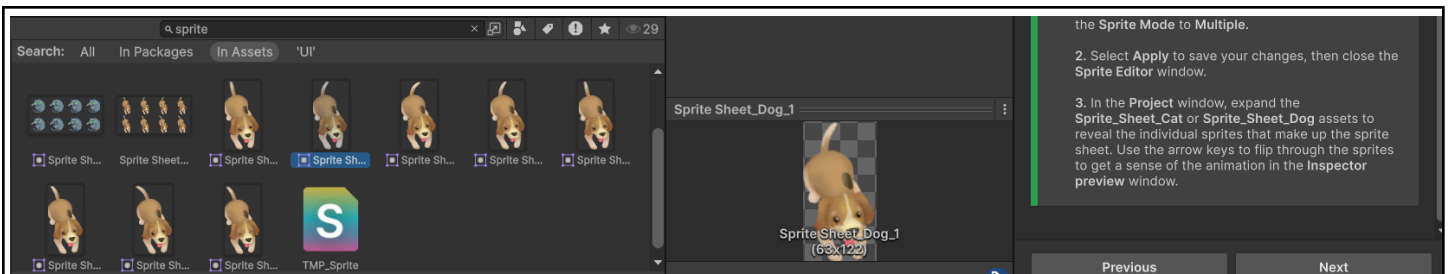




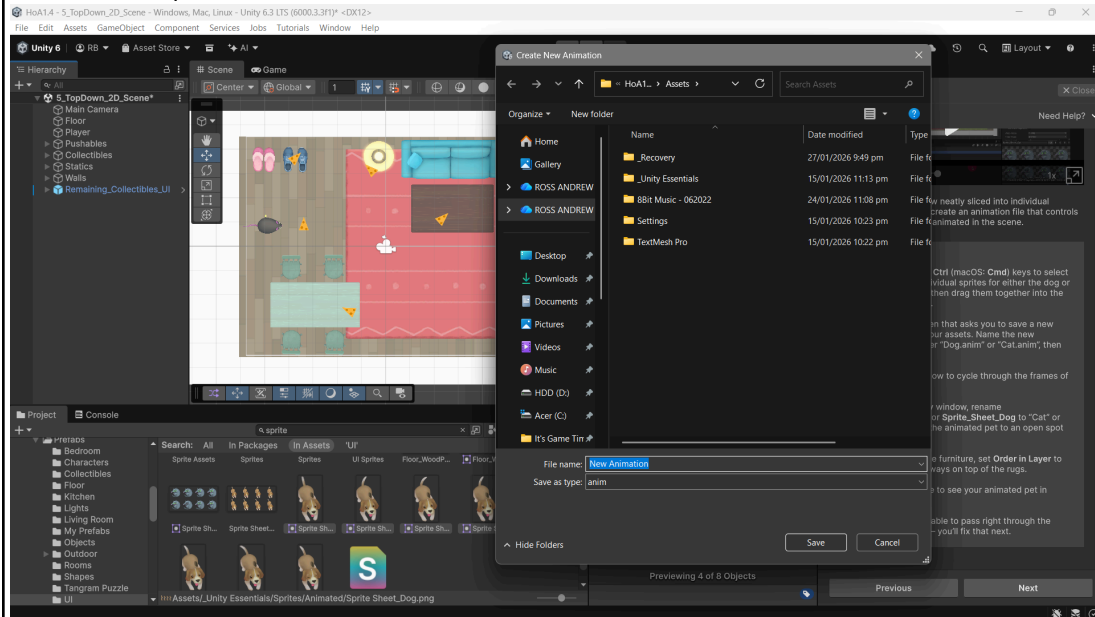
Checked the animated 2D dog sprite then changed the sprite mode to multiple as instructed



Sliced the sprite by following the instructions: Keep the type automatic, then say yes even if a warning msg tells me the possibility of sprite data loss (I do not understand for now, but i just ticked yes as instructed), then saved i



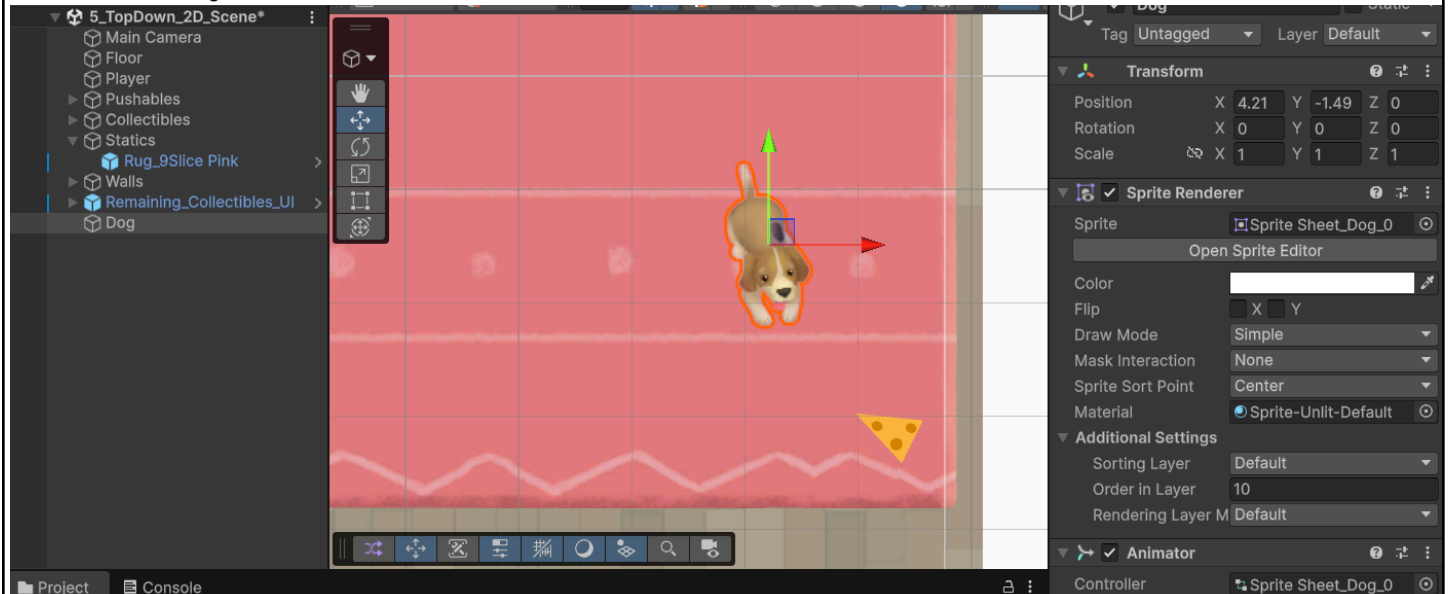
The individual sprites of the animation is now visible



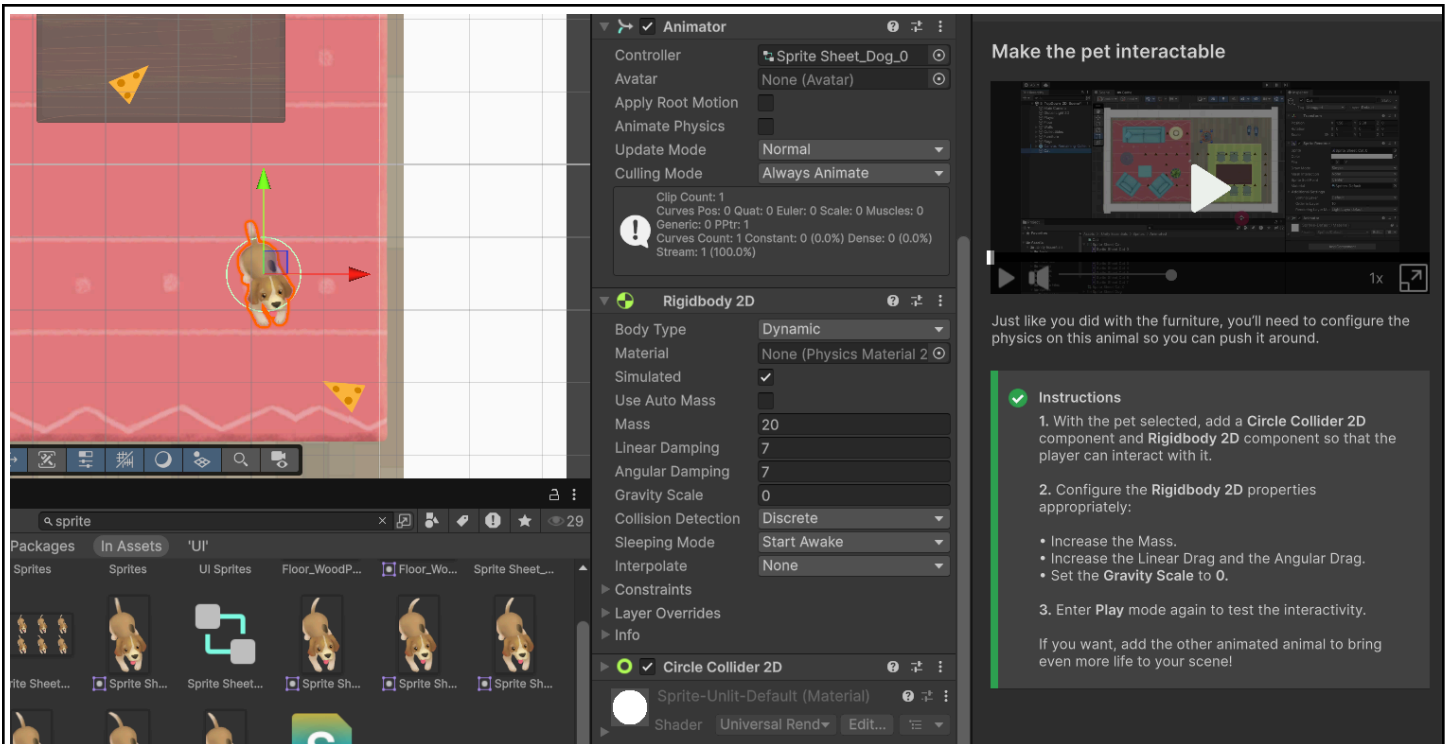
Dragged all of the individual sprites to the hierarchy window in which a file explorer window was opened, prompting me to save each sprite as a combined .anim file



I named the file Dog.anim



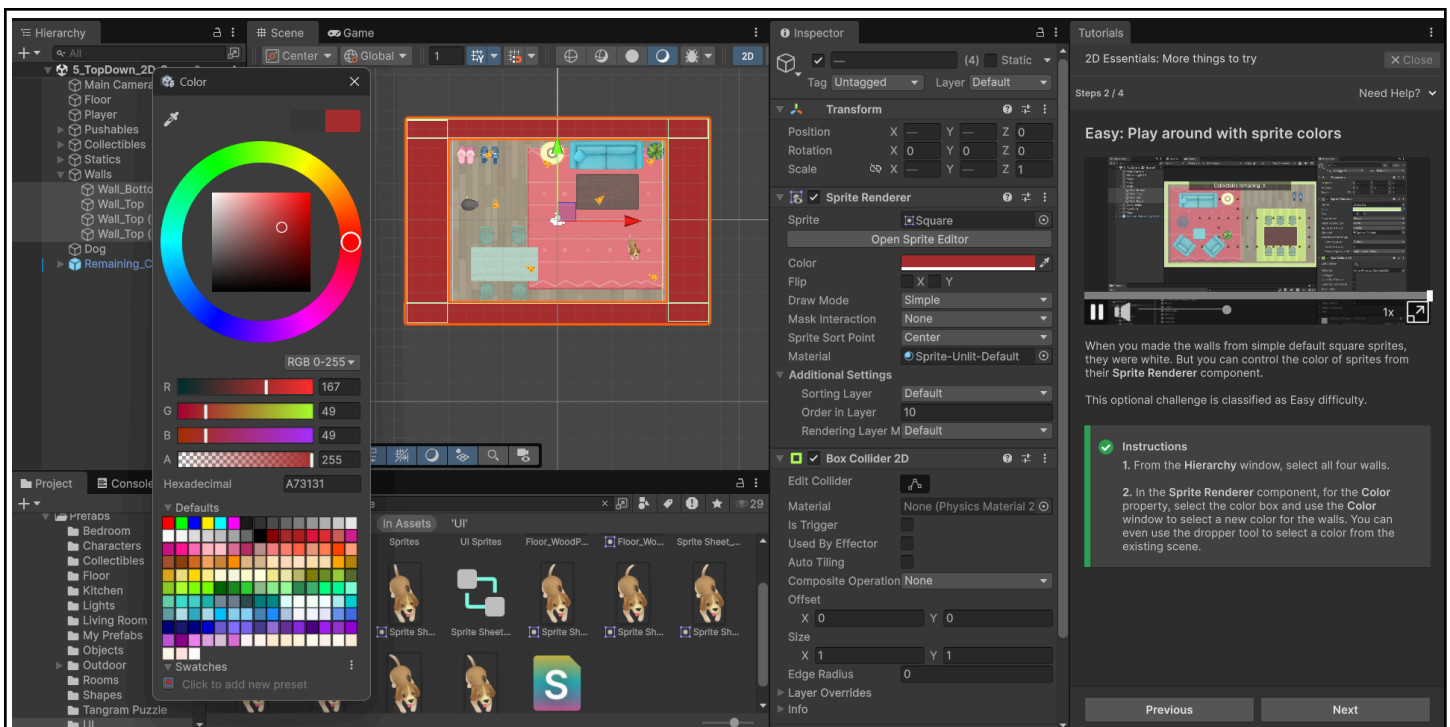
Moved the dog to my desired place, then changed the layer to 10; the same layer as the "Pushables"



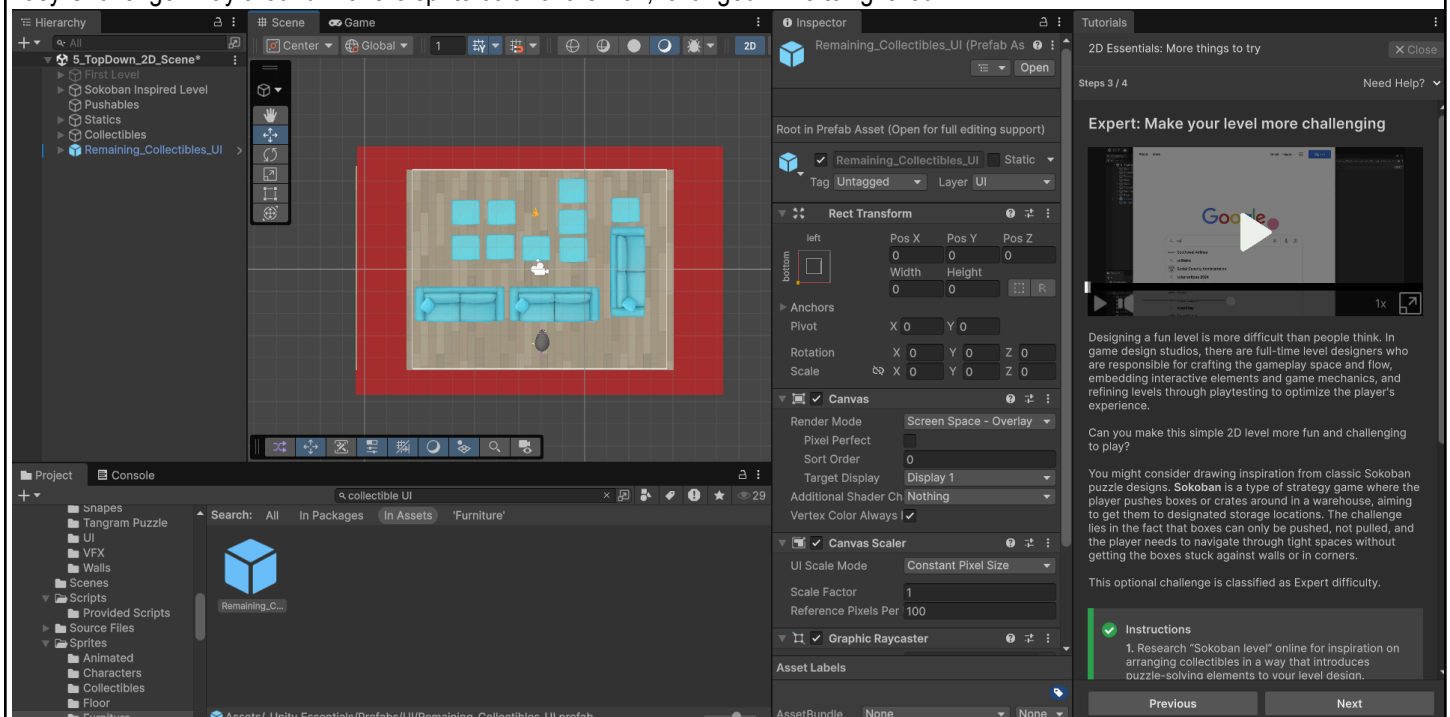
Made the dog interactable by adding a circle collider and a 2D rigid body then setting some parameters such as the linear and angular damping as well as the gravity scale to match the other GameObjects



I am successful in making the dog interactable



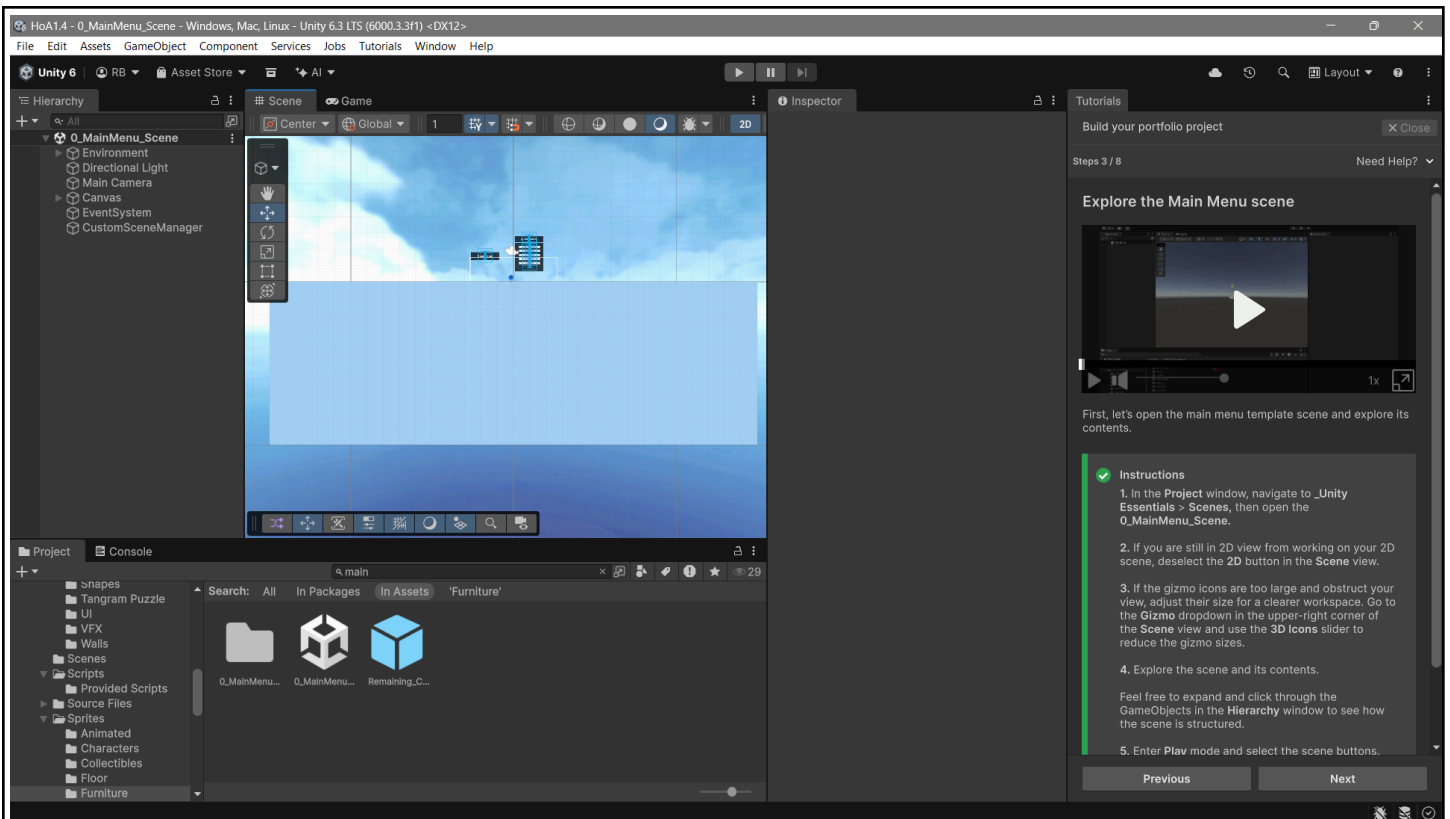
Easy Challenge: Play around with the sprite color of the wall, i changed mine to light red



Expert challenge: Make my level more challenging. I took a maze approach and made a simple maze that even a 5 year old can complete. It is not challenging per se, but I did make my own level. Perfect for a "maze" type game that features harder and harder progressive levels; and the one I made is the tutorial. *I did not make anything too hard, just demonstrated the skills I've learned throughout the tutorial.*

Check out the demonstration here: [MyOwn2DLevelDemo.mp4](#)

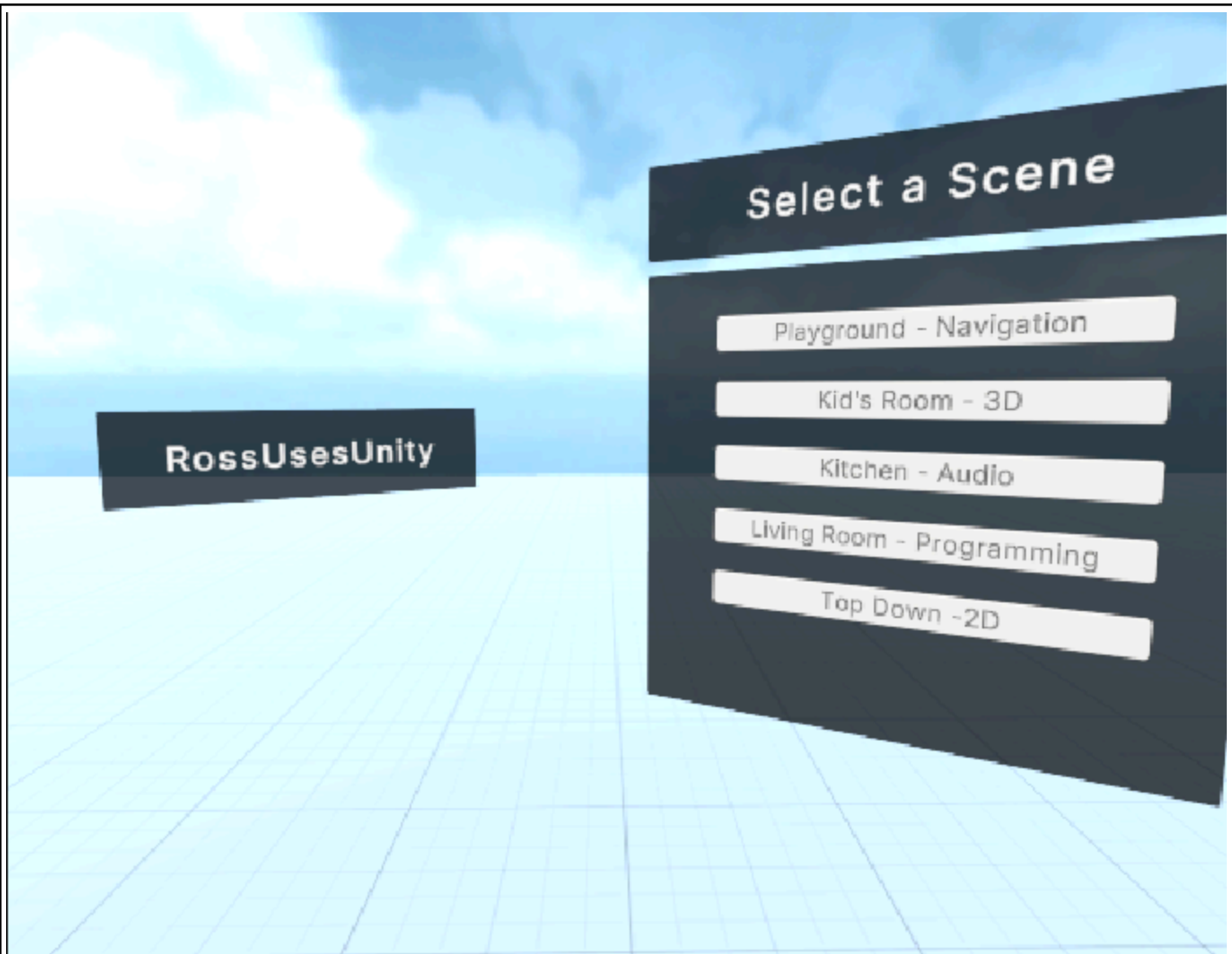
PUBLISHING ESSENTIALS



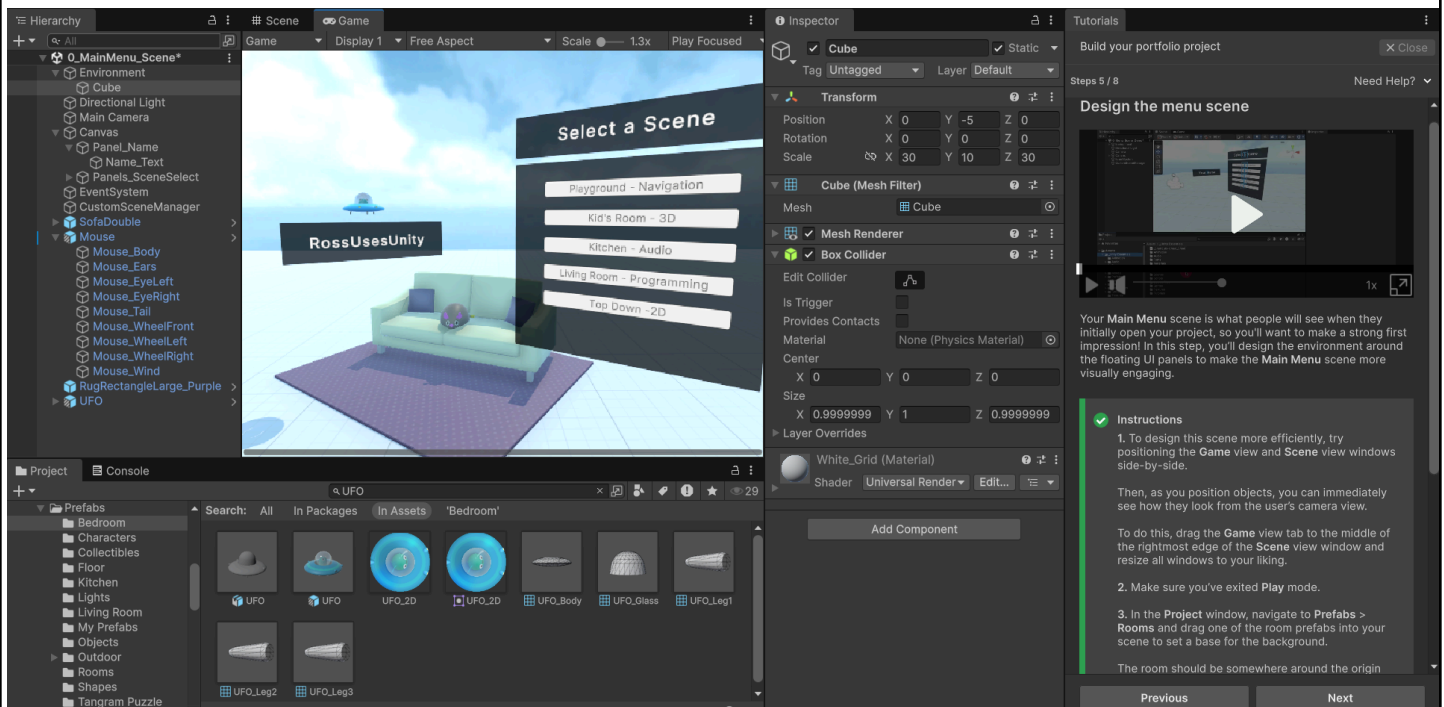
Explored through the main menu scene provided by Unity. For now, none of the buttons work



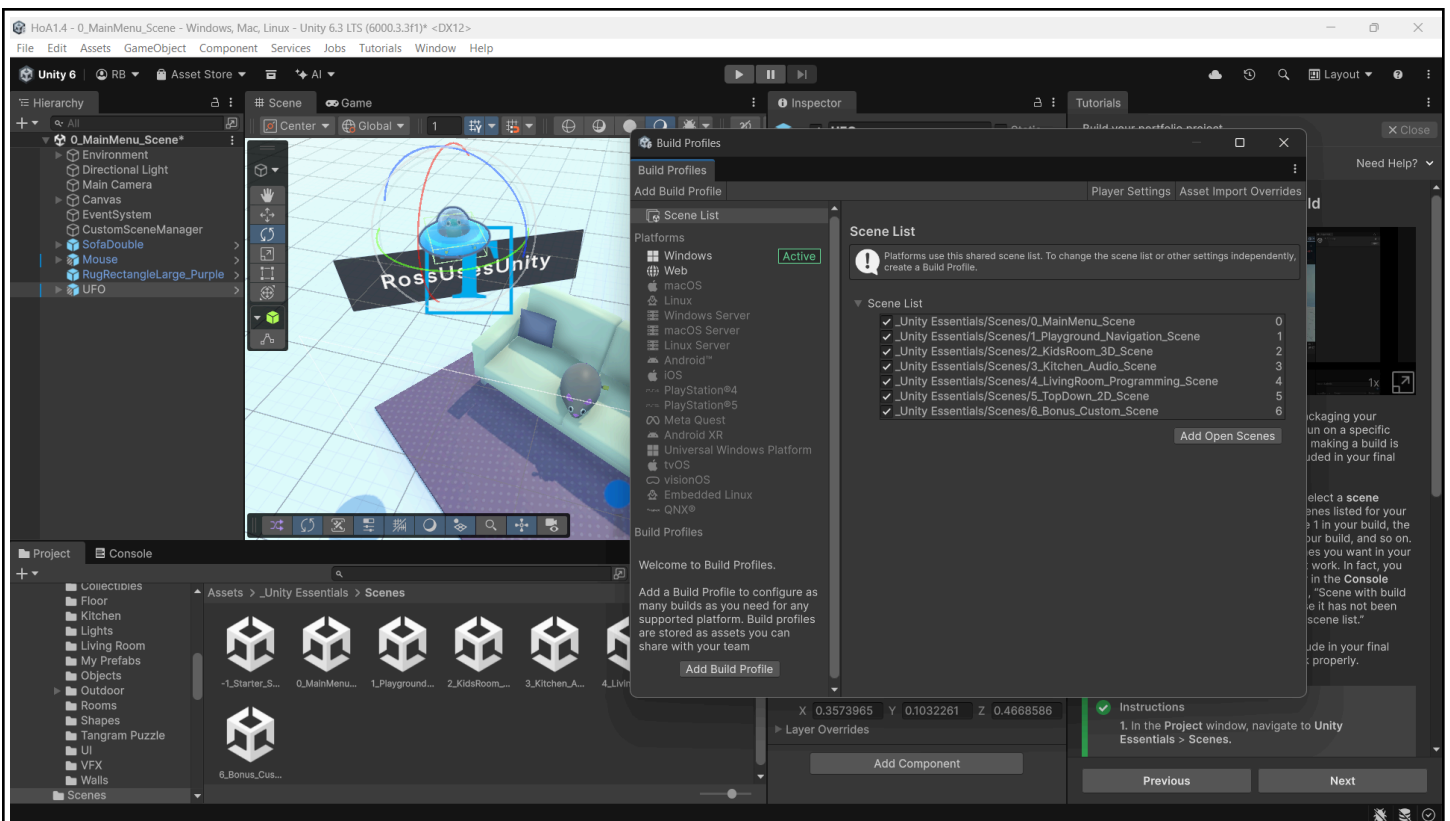
Added my name to the TextInput field in the Canvas Renderer



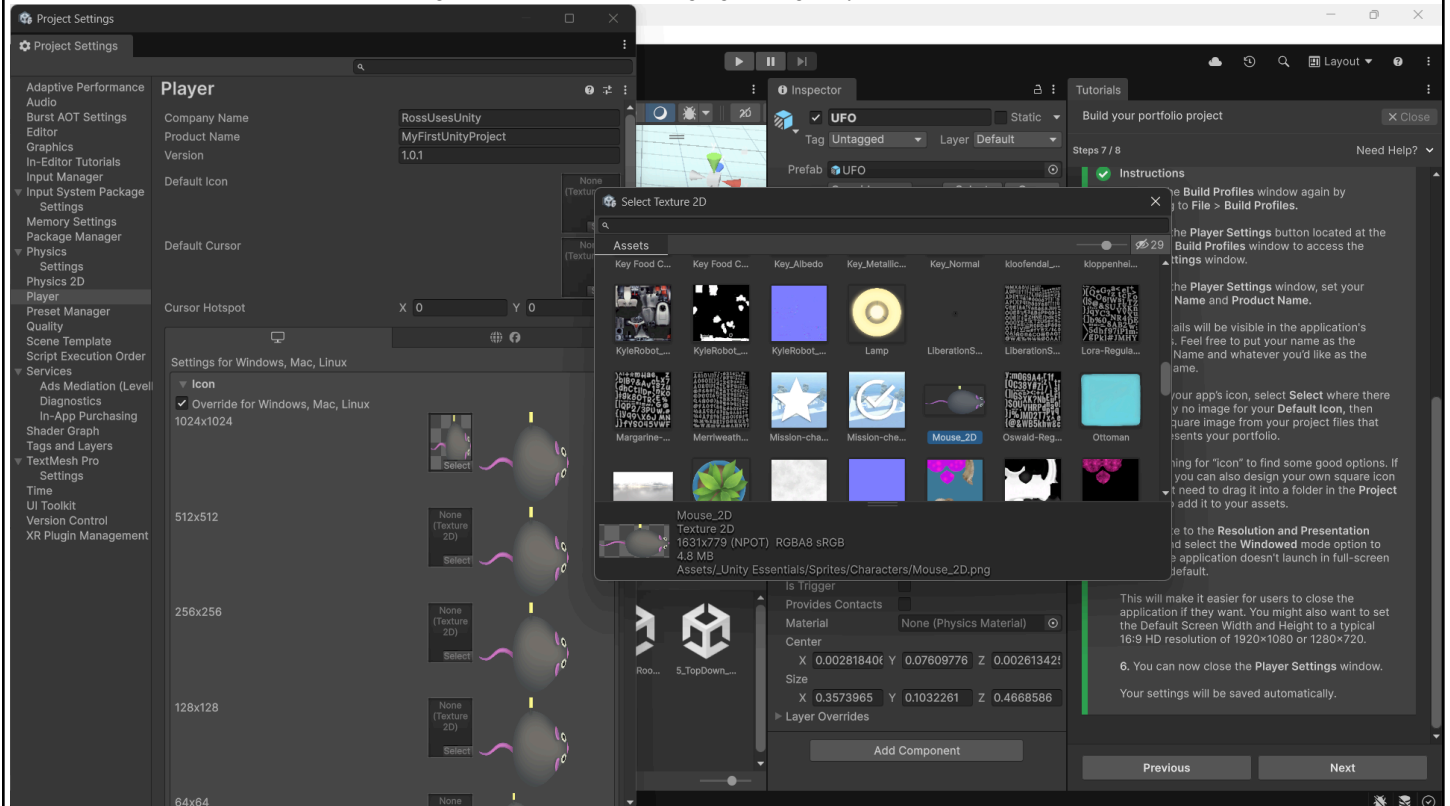
It is now reflected in the scene



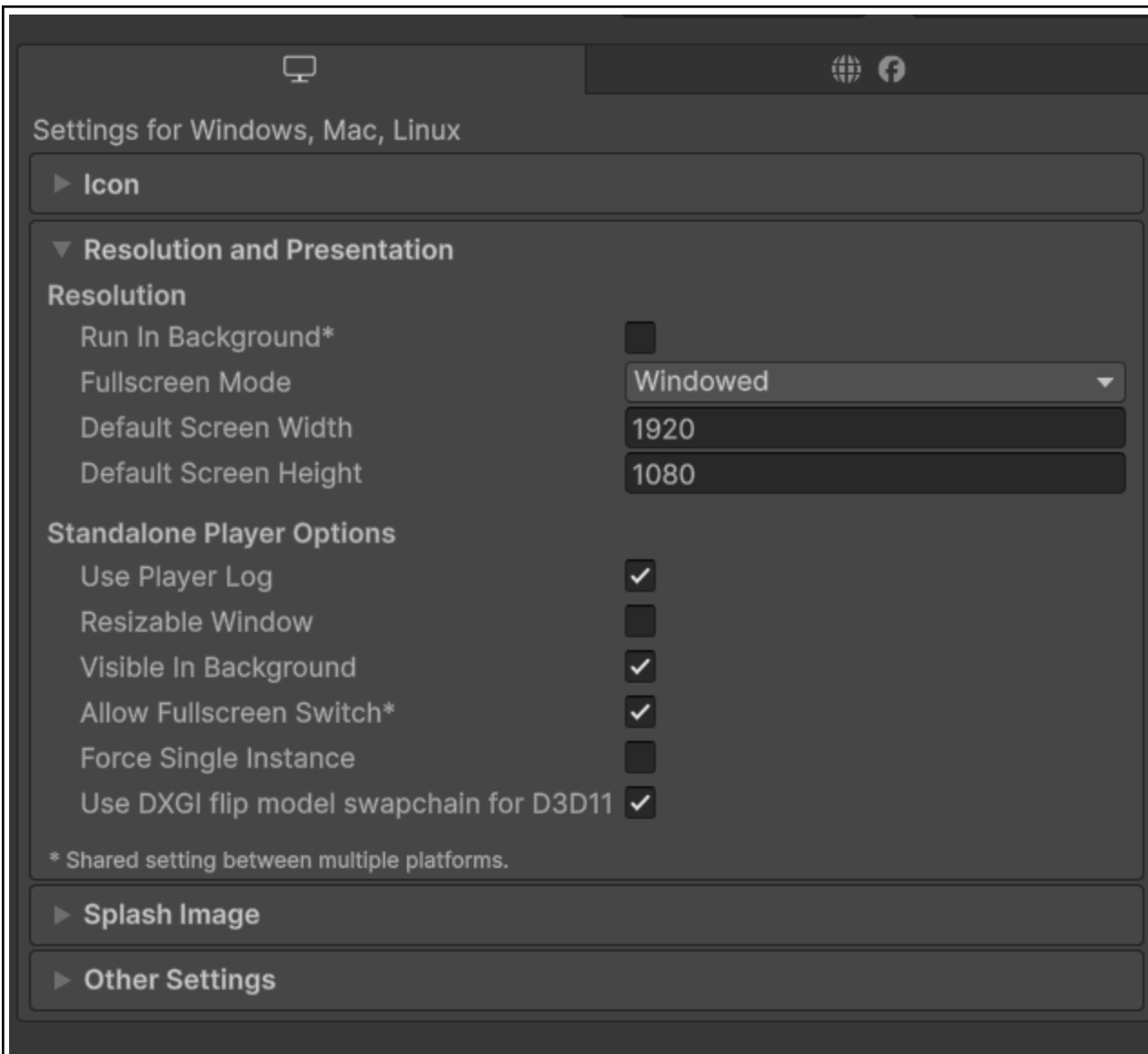
I added my own personal design to the menu scene, featuring the mouse that carried me throughout my entire journey.



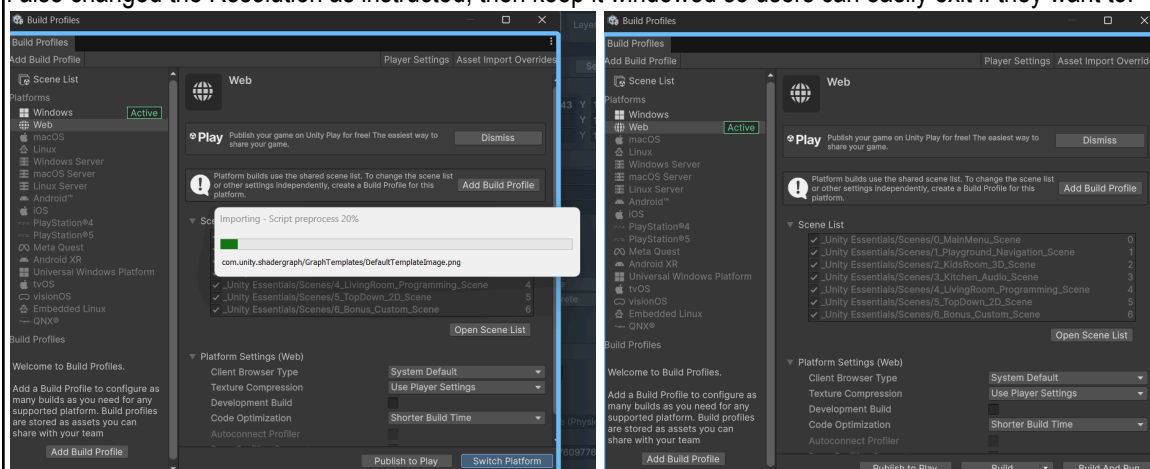
I added my scenes to the build profile as instructed. I can now navigate to each and every one of my own scenes that I've built for the past couple Hands-on Activities. I feel a great sense of pride surging through my veins.



I then changed my company name to my Unity Hub username, as well as the Project name to reflect my first project. I made a mouse as my icon, as this mouse is my "hard carry" throughout my entire navigation of Unity.



I also changed the Resolution as instructed, then keep it windowed so users can easily exit if they want to.



Selected WebGL as my platform



WebBuilds

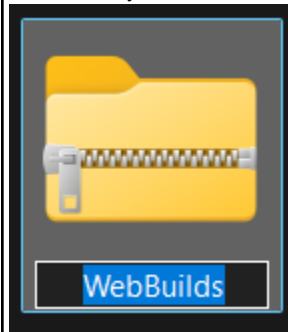
Importing - Compress 50% (busy for 32s)...

Resources/PerformanceTestRunInfo.json

Made a WebBuilds folder as instructed then used that folder as my build location

 Build	31/01/2026 11:14 pm	File folder	
 TemplateData	31/01/2026 11:14 pm	File folder	
 index	31/01/2026 11:14 pm	Brave HTML Docu...	6 KB

Successfully built



Compressed my build as instructed

[Play](#) [Showcases](#) [Struckd Web](#) [Struckd Mobile](#) [Exporter](#) [Learn](#) [FAQ](#)

[Play](#)

MyFirstUnityProject

0 plays

 RossUsesUnity

This is the WebGL build of my first Unity project, developed for the subject CPE 411 – Interactive Extended Reality under the Bachelor of Science in Computer Engineering program at T.I.P

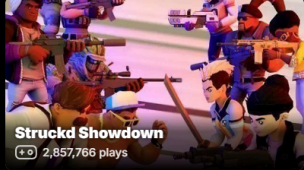


0 / 200

Made with

 [Download](#)

You may also like



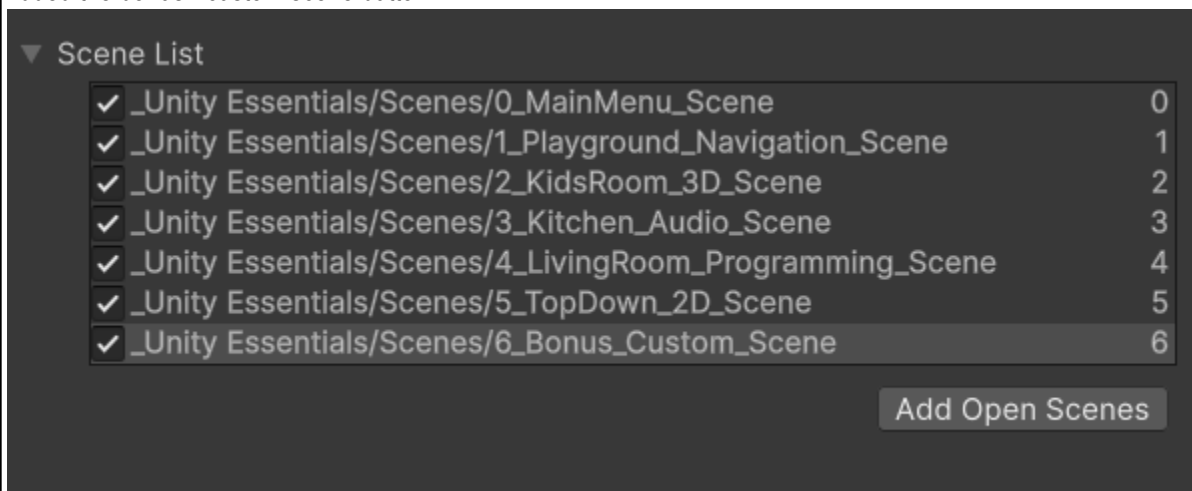
Struckd Showdown
2,857,768 plays

<https://play.unity.com/en/user/c3f81d4a-4629-477b-8175-dcfd6e439ca3>

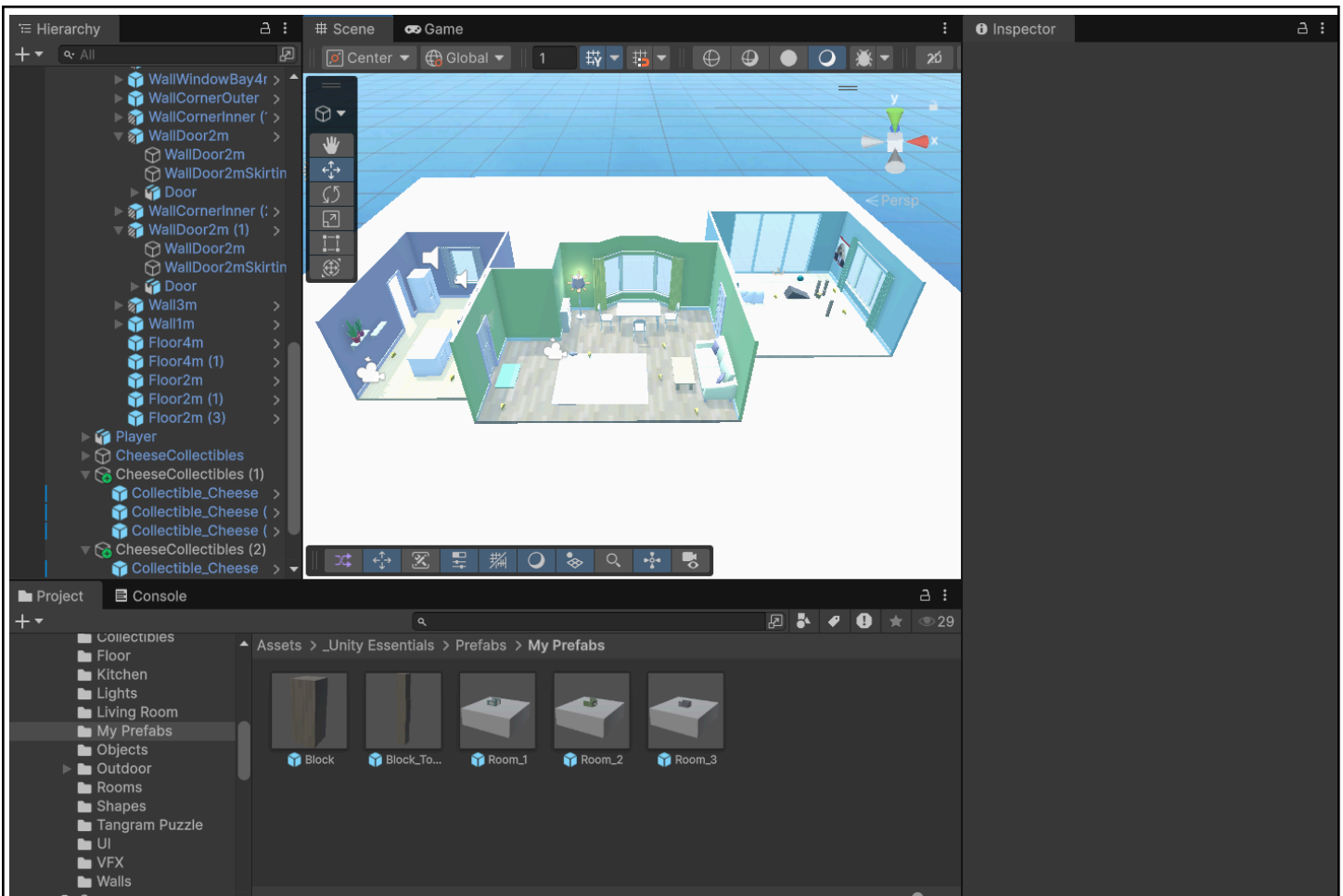
Successfully uploaded my build to WebGL



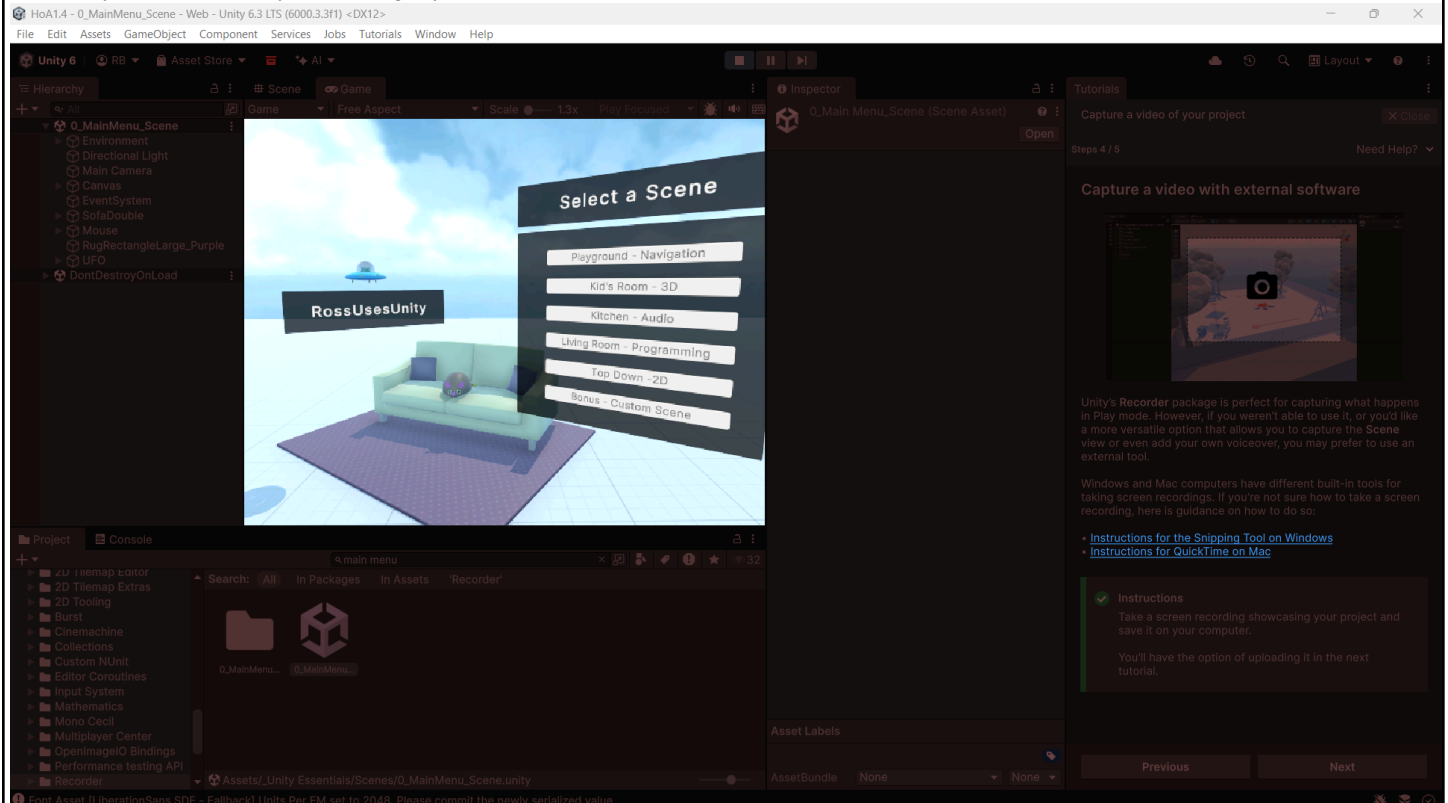
Added the bonus - custom scene button



Added the Bonus Custom Scene to the scene list in the build profile



Added my bonus scene by combining my previous three rooms, added more collectibles to promote exploration



Recorded a video using my own Snipping Tool

4. Output

Your results

Congratulations! You passed. If you would like to score higher, you can [retake](#) the quiz. We always save your highest score.

Passed • You got 5 of 5 correct

+60 XP

You gained 60 XP points towards
Unity Essentials

You completed this quiz on January 31, 2026

Next up in Unity Essentials: In-Editor



Publishing Essentials: In-Editor

+600 XP • 2 Hour 25 Mins

[Continue to next](#)

COMPLETED ✓

5 | 2D Essentials

Learn how developing for 2D differs from 3D as you create a 2D version of the living room scene.

[→](#)

COMPLETED ✓

6 | Publishing Essentials

Combine all of your skills and all of your scenes together into one cohesive portfolio project that you can share with others.

[→](#)

My Build Link: <https://play.unity.com/en/games/1f524480-600b-49db-8ea8-216f484ce75c/myfirstunityproject>

5. Supplementary Activity

No supplementary activity was given.

6. Assessment Rubric

Lab Activity Rubric							 
Criteria		Ratings					Pts
 SO 7 PI 1 Student Outcome 7.1 Acquire and apply new knowledge from outside sources. threshold: 4.8 pts	6 pts Excellent Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently and applies knowledge learned into practice	5 pts Good Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently	4 pts Satisfactory Look beyond classroom requirements, showing interest in pursuing knowledge independently	3 pts Unsatisfactory Begins to look beyond classroom requirements, showing interest in pursuing knowledge independently	2 pts Poor Relies on classroom instruction only	1 pts Very Poor No initiative or interest in acquiring new knowledge	6 pts
 SO 7 PI 2 Student Outcome 7.2 Learn independently threshold: 4.8 pts	6 pts Excellent Completes an assigned task independently and practices continuous improvement	5 pts Good Completes an assigned task without supervision or guidance	4 pts Satisfactory Requires minimal guidance to complete an assigned task	3 pts Unsatisfactory Requires detailed or step-by-step instructions to complete a task	2 pts Poor Shows little interest to complete a task independently	1 pts Very Poor No interest to complete a task independently	6 pts
 SO 7 PI 3 Student Outcome 7.3 Critical thinking in the broadest context of technological change threshold: 4.8 pts	6 pts Excellent Synthesizes and integrates information from a variety of sources; formulates a clear and precise perspective; draws appropriate conclusions	5 pts Good Evaluate information from a variety of sources; formulates a clear and precise perspective.	4 pts Satisfactory Analyze information from a variety of sources; formulates a clear and precise perspective.	3 pts Unsatisfactory Apply the gathered information to formulate the problem	2 pts Poor Gather and summarized the information from a variety of sources but failed to formulate the problem	1 pts Very Poor Gather information from a variety of sources	6 pts
 SO 7 PI 4 Student Outcome 7.4 Creativity and adaptability to new and emerging technologies threshold: 4.8 pts	6 pts Excellent Ideas are combined in original and creative ways in line with the new and emerging technology trends to solve a problem or address an issue.	5 pts Good Ideas are creative and adapt the new knowledge to solve a problem or address an issue	4 pts Satisfactory Ideas are creative in solving a problem, or address an issue	3 pts Unsatisfactory Shows some creative ways to solve the problem	2 pts Poor Shows initiative and attempt to develop creative ideas to solve the problem	1 pts Very Poor Ideas are copied or restated from the sources consulted	6 pts
Total Points: 24							